



SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNICAL UNIVERSITY OF MUNICH

Bachelor's Thesis in Information Systems

**Exploring Human-AI Collaboration in
Counterspeech Generation: Development of
a Browser Extension to Support Users in
Responding to Online Hate Speech**

Tammy Kathrina Anja Schmidt





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**Erforschung von Mensch-KI-Zusammenarbeit bei der
Verfassung von Gegenrede: Entwicklung einer
Browser-Erweiterung zur Unterstützung beim Antworten auf
Online-Hassrede**

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I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 13/04/2026

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Abstract

This thesis addresses the persistent challenge of online hate speech (HS) by investigating the generation of counterspeech (CS) as a constructive alternative to traditional moderation approaches. It explores human–AI collaboration (HAIC) as a middle ground between fully manual writing, which requires substantial skill and effort, and fully automated systems, which often lack authenticity and meaningful human involvement. To this end, the design, implementation, and evaluation of an AI-powered browser extension to support CS writing is presented. The system leverages a large language model which is tailored to the task through prompt engineering and in-context learning, using similarity-based retrieved examples from the Multitarget CONAN dataset. The study examines how such a system can seamlessly assist users in composing authentic and effective CS while also assessing its impact on time requirements. Central to this evaluation, the thesis proposes a novel framework to assess the authenticity of AI-assisted CS across five dimensions.

Results from a user study indicate that a browser extension with an integrated pop-up interface is a promising design choice for enabling seamless writing support, while the interactive features – such as allowing users to provide input and select from multiple generated suggestions – help preserve a sense of personal authenticity. The findings underscore the importance of designing CS systems that foster active user engagement rather than relying on one-click automation that minimizes human involvement. Furthermore, while the use of the browser extension increased the time spent to compose responses, it led to measurable improvements in the effectiveness of the generated CS. Insights from the user study highlight the need for greater stylistic diversity and more transparency in system functioning, among other further refinements. The thesis concludes with a critical reflection on its limitations and outlines directions for future research. Overall, it provides novel insights into HAIC for CS generation and contributes to the development of effective and user-centered tools to combat online HS.

Diese Abschlussarbeit beschäftigt sich mit der anhaltenden Herausforderung von Online-Hassrede, indem sie die Generierung von Gegenrede (sogenannte Counter-speech (CS)) als konstruktive Alternative zu klassischen Moderationsansätzen untersucht. Im Fokus steht dabei Mensch-KI-Zusammenarbeit als Mittelweg zwischen vollständig manuellem Schreiben, das viel Zeit und Können erfordert, und vollständig automatisierten Systemen, denen es oft an Authentizität und menschlicher Beteiligung fehlt. Zu diesem Zweck werden Design, Implementierung und Evaluation einer KI-gestützten Browser-Erweiterung vorgestellt, die beim Verfassen von CS unterstützt. Das System basiert auf einem Large Language Model, das mithilfe von Prompt Engineering und In-Context Learning gezielt auf die Aufgabe angepasst wird, wobei ähnlichkeitsbasierte Beispiele aus dem Multitarget-CONAN-Datensatz zum Einsatz kommen. Die Studie untersucht, inwiefern ein solches System Nutzende unterstützen kann, authentische und wirksame CS zu verfassen, und betrachtet zugleich den Einfluss auf den benötigten Zeitaufwand. Im Rahmen der Evaluation wird zudem ein neu entwickeltes Framework vorgestellt, das die Authentizität KI-unterstützter CS entlang von fünf Dimensionen bewertet.

Die Ergebnisse einer Nutzerstudie zeigen, dass die Browser-Erweiterung mit integriertem Pop-up-Fenster eine vielversprechende Lösung für eine nahtlose Schreibunterstützung darstellt. Gleichzeitig tragen interaktive Funktionen – etwa die Möglichkeit, eigene Inhalte einzubringen und zwischen mehreren Vorschlägen zu wählen – dazu bei, ein Gefühl persönlicher Authentizität zu erhalten. Die Ergebnisse unterstreichen, wie wichtig es ist, CS-Systeme so zu gestalten, dass sie aktive Beteiligung fördern, anstatt auf vollständige Automatisierung zu setzen. Darüber hinaus zeigt sich, dass der Einsatz der Browser-Erweiterung zu einem erhöhten Zeitaufwand führt, gleichzeitig aber messbare Verbesserungen in der Wirksamkeit von CS erzielt werden. Erkenntnisse aus der Nutzerstudie machen zudem deutlich, dass es – neben weiteren Verbesserungsmöglichkeiten – insbesondere mehr stilistische Vielfalt und größere Transparenz in der Funktionsweise des Systems braucht. Abschließend reflektiert die Arbeit kritisch ihre Limitationen und skizziert Ansätze für zukünftige Forschung. Insgesamt liefert sie neue Einblicke in Mensch-KI-Zusammenarbeit für die Generierung von CS und leistet einen Beitrag zur Entwicklung effektiver und nutzerzentrierter Werkzeuge zur Bekämpfung von Online-Hassrede.

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1 Introduction

While internet platforms offer powerful opportunities for connection and virtual communication, they have also emerged as playgrounds for online hostility. Most internet users have likely encountered hate speech (HS) within comment sections or social media feeds, making it a familiar aspect of the modern digital experience. The United Nations Strategy and Plan of Action on Hate Speech defines HS as “any kind of communication in speech, writing or behaviour, that attacks or uses pejorative or discriminatory language with reference to a person or a group on the basis of who they are, in other words, based on their religion, ethnicity, nationality, race, colour, descent, gender or other identity factor” [1, p. 8]. The aim of HS is to harm, dehumanize, intimidate, degrade, and victimize its targets, while fostering a broader climate of insensitivity and hostility toward them [2]. Although hateful online content takes many nuanced forms and is described using a range of terms in the literature, this study adopts the widely recognized and accessible term HS. In doing so, it encompasses various forms of abusive online language, including implicit or covert toxicity [3]–[5], microaggressions [6], and stereotyping [7].

Across all degrees of hate intensity, HS can have serious adverse effects on the mental health and sense of security of both victims and bystanders, contributing to reduced self-esteem, stress, anxiety, depression, panic attacks, and, in extreme cases, even suicidal thoughts [8]–[10]. Beyond individual impacts, HS can also have broader societal consequences, fostering processes such as radicalization and polarization [11]. Given these risks, there is a strong societal interest in developing effective strategies to counter online HS and promote safer digital environments [12], [13].

One of the most widely used mitigation strategies is content removal, including deleting posts or comments as well as suspending or shadow-banning user profiles. However, such moderation practices often challenge the definition of free speech and are frequently criticized as unjustified censorship [14], [15]. Moreover, factors such as cultural differences, biased annotators or biased algorithms can lead to over-blocking of content [16], [17]. At the same time, HS expressed in more subtle forms may remain unaddressed because it is not detected [4], [5] or does not necessarily violate existing community guidelines [3]. Finally, granting social media platforms the authority to moderate and intervene in online communication raises additional concerns about an unprecedented level of centralized control [18].

In this context, counterspeech (CS) represents a promising alternative, grounded in a philosophy that dates back to 1927 with Justice Brandeis’ principle of more speech, which advocates addressing hate through dialogue and education rather than enforced silence [19]. Most definitions of CS draw on the work of Benesch [20] and Schieb *et al.* [21], who characterize it as non-aggressive textual feedback that challenges harmful narratives by offering counterarguments or alternative viewpoints. Without reinforcing hostility, CS seeks to express support for victims, encourage authors to reconsider their statements, and critically deconstruct problematic content for bystanders [3]. Moreover, setting an example of constructive and civil engagement through CS has been shown to foster a more respectful tone in subsequent discussions [21]–[25]. As such, CS offers a compelling complement or alternative to deletion-based moderation strategies, with the potential to promote a positive cultural shift through dialogue [26].

Despite its potential, many users in practice refrain from engaging in CS, partly due to the considerable challenges involved in formulating appropriate responses [27]. A range of barriers discourages participation, including time constraints, emotional strain, limited skills, fear of being unqualified, concerns about personal repercussions, and uncertainty about the impact of such interventions [26], [28].

Given these obstacles, as well as the sheer scale of online HS, artificial intelligence (AI) holds significant promise for supporting and facilitating CS writing. Recent research has extensively examined the potential of large language models (LLM) for automatic generation of CS, demonstrating their strong capabilities in natural language processing and generation. In particular, by drawing on patterns learned from large bodies of data, LLMs can support the structuring of arguments in a more coherent manner, help identify appropriate and nuanced wording, and include insights derived from prior research, thereby contributing to more effective responses.

However, despite their strong performance, LLMs remain susceptible to issues such as generating abusive content, misinterpretations, or hallucinations. In this context, it is important to recognize that inappropriate CS may inadvertently exacerbate online hate [29]. Furthermore, fully AI-generated CS often lacks the personal voice, emotional authenticity, and contextual nuance that are essential for effective real-world interactions [30]. To ensure quality and ethical integrity, there is broad consensus that AI systems should be deployed as assistive suggestion tools under human supervision rather than as fully autonomous agents [31]–[34]. As a result, there is a growing need for human–AI collaboration (HAIC) tools that combine the generative capabilities of LLMs with user-friendly and accessible interfaces.

Addressing this need, this bachelor’s thesis developed and evaluated an AI-powered browser extension as a CS writing assistant, introducing a novel approach to delivering AI support directly within the user’s browsing context. Prompt engineering, including in-context learning (ICL) with example retrieval from the Multitarget CONAN dataset,

is employed to adapt the underlying model to the task of CS generation. The extension is further designed with the aim of supporting users in staying actively engaged in the generation process, thereby helping to preserve their own voice and fostering a sense of ownership over the co-written CS [26], [28], [35]. In this context, this bachelor's thesis aims to answer the following research question:

When exposed to online HS, how can an AI-powered browser extension *seamlessly* support users in composing *authentic* and *effective* CS, and what is its impact on *time* requirement?

The following work is structured as follows: Chapter 2 provides an overview of the related research, including key findings on what constitutes effective CS, the state of the art in automatic CS generation – covering different model adaptation strategies and the rationale for selecting prompt engineering with ICL – as well as existing tools addressing online HS, thereby highlighting the novelty of a browser-based CS writing assistant. Chapter 3 then presents the system design within the design space that guided the development, introducing the ethical framework and the overall system architecture. The concrete implementation of each component is detailed in Chapter 4. The developed extension is subsequently evaluated through a user study, with the experimental setup, evaluation design, and results across the four dimensions of interest – seamless integration, perceived authenticity, effectiveness of the generated output, and time requirements – reported in Chapter 5. Finally, the findings are critically discussed in Chapter 6 and the thesis is concluded in Chapter 7.

2 Related Work

This chapter presents an overview of the relevant literature. It (1) examines key insights into what makes CS effective, (2) reviews current approaches to automatic CS generation – including different model adaptation strategies and the choice of prompt engineering with ICL – and (3) discusses existing tools for addressing online HS.

2.1 Effectiveness of Counterspeech

Understanding what makes CS effective first requires defining effectiveness itself. The effectiveness of CS can be understood along several dimensions, including its ability to change the HS writer’s stance, de-escalate conflict, make victims feel supported, encourage bystanders to intervene, and ultimately improve the overall tone of online discourse [3], [29], [36].

While the effects on HS authors and targets remain uncertain, the impact on bystanders has been more extensively and successfully studied. CS can be seen as a form of online civility, which describes a prosocial behavior marked by mutual respect, politeness, and topic relevance [24]. Research on contagious civility and pro-social example setting consistently shows that users tend to mirror the tone and behavior modeled by others, indicating that online conversations tend to strongly shift in line with prior comments. In other words, exposure to civil or thoughtful comments increases the likelihood that subsequent participants will also respond in a constructive manner, suggesting that visible examples of respectful dialogue can shape the overall rhetoric of online conversations [21]–[24], [37], [38]. This seems to be especially true if the CS comes from a high-status ingroup member [25].

Nevertheless, research has shown that not all CS strategies equally contribute to the above-mentioned goals, and their success depends on the context and target audience [39]. Studies have identified several favorable strategies, such as empathy [3], [7], [29], [40], [41], positivity [3], [42], disapproval [41], social norms [41], or warning of consequence [7], [29]. In contrast, moralizing or hostile forms of CS are likely to provoke defensive or aggressive reactions, increasing hate intensity [3], [29], [39], [43]. The effectiveness of questioning and denouncing shows different degrees among studies indicating that their impact depends on context and tone [7], [42]. Use of humor in CS has shown mixed results, with some studies reporting positive [29], [42] and

others negative effects [7], suggesting that humorous interventions should be applied with caution. A similar ambiguity applies to fact-checking strategies: sometimes they increase hate among users [29], [42], [44], other times they can be effective in supporting victims and preventing them from hating back [45].

It is important to note that the effectiveness of CS remains challenging to measure, as its effects are often invisible, difficult to observe in real-world settings, or only emerge over the long term. This difficulty is further compounded by the context-dependent nature of CS, the diversity of audiences and interpretations, and the challenge of isolating its impact from other influencing factors in dynamic online environments [36], [46]. Further empirical work is needed to better understand the nuanced impact of different strategies.

Despite these limitations, there is strong consensus that empathetic and positive tones are generally more effective than hostile ones. Baider [3], who analysed 15,000 real-world comments from Twitter and YouTube on CS strategies and their effects on the community, provides a clear recommendation to focus on constructive and positive approaches. At the same time, prior research supposes that effective CS should also exhibit common quality attributes, such as specificity, contextual relevance, or clear structure in order to be persuasive.

However, many users are unaware of these strategic differences or struggle with negative emotions and difficulties in articulating their thoughts when encountering HS [28]. AI assistance bears the potential to reduce this emotional burden and to craft measured and well-formulated responses [7].

Consequently, the proposed browser extension aims to support users in composing CS that aligns with research-based principles of effectiveness. How effectiveness is evaluated within this study is explained in Section 5.4 in detail.

2.2 Automatic Counterspeech Generation

In line with the broader literature on adapting LLMs to task-specific applications [47], research on automated CS generation has primarily explored three main strategies: fine-tuning [32], [48]–[53], prompt engineering [6], [49], [51], [54], and retrieval-augmented generation (RAG) [32], [55]. While these approaches differ in how they adapt models to the task, they all aim to improve the quality and relevance of generated responses. Fine-tuning enables models to internalize stylistic and tonal patterns from task-specific training data. In contrast, prompt engineering, particularly through few-shot prompting or ICL, guides model behavior by providing representative examples directly within the prompt. RAG, on the other hand, enhances outputs by connecting the model to external knowledge sources, allowing it to incorporate up-to-date and contextually

relevant information. Among these approaches, ICL and RAG have gained increasing attention due to their flexibility and lower computational requirements compared to fine-tuning [47].

In the context of this work, the primary objective is not to generate fact-based arguments, but to support the formulation of well-phrased, contextually appropriate, and stylistically effective CS. As such, approaches that emphasize linguistic adaptation – namely fine-tuning and ICL – are particularly relevant. Prior research suggests that there is no universally superior method: Alghisi *et al.* [56] show that the effectiveness of fine-tuning versus ICL depends on both the underlying model and the dialogue type. However, in the specific domain of CS generation, prompting large pre-trained models has been shown to outperform smaller fine-tuned models [49], [51].

More broadly, the strong ICL capabilities of modern LLMs have contributed to a shift toward prompt-based approaches. Y. Qian *et al.* [57] demonstrate that ICL is highly effective for empathetic dialogue generation, particularly when examples are selected based on semantic similarity using sentence embeddings. While CS generation involves additional complexity, these insights can, to some extent, be transferred to CS, as both tasks share key characteristics. In particular, effective CS often relies on understanding emotions and responding in a constructive and empathetic manner [3], [7], [29], [40], [41].

Building on these findings, Ashida *et al.* [6] generated CS using ICL with a small set of static examples from the Multitarget CONAN dataset, which is among the gold standard of HS/CS datasets (see Subsection 4.3.1). In their approach, the same five examples are reused consistently across all prompts. The present work extends this method by dynamically retrieving different examples for each prompt based on semantic similarity to the input HS instance. This is achieved by querying a vector database of stored examples, thereby leveraging the strengths of RAG. This dynamic selection enables a closer alignment between the input and the provided examples, increasing contextual relevance and is expected to improve generation quality. This expectation is supported by J. Liu *et al.* [58], who show that similarity-based example retrieval significantly enhances ICL performance across a range of natural language generation tasks.

Overall, the combination of retrieval-augmented prompt engineering with ICL represents a state-of-the-art and highly promising approach, making it particularly well-suited for adapting the underlying model of the browser extension to the CS generation task.

2.3 Tools Against Hate Speech

Much of the existing CS research focuses on optimizing generation models for fully automated use, paying limited attention to human–AI interaction. However, to ensure quality and ethical integrity, there is broad consensus that AI-based CS generators should be deployed as assistive suggestion tools under human supervision rather than as fully autonomous agents [31]–[34]. Bridging the research gap between model development and deployment through user-centered tools is therefore essential to align technical progress with users’ needs and to enable meaningful practical impact.

In 2024, Mun *et al.* [26] conducted in-depth interviews with 10 experienced counterspeakers alongside a large-scale survey of 342 everyday social media users to investigate how AI can support users in formulating CS. Their findings highlight three key potentials of AI: (1) empowering users to compose CS they perceive as impactful, (2) facilitating more empathetic and emotionally intelligent responses, and (3) retrieving relevant information for fact-based and well-reasoned arguments. Among these aspects, usability emerges as a central requirement, particularly seamless integration into existing workflows to increase efficiency. One participant in their study explicitly expressed a preference for remaining within the same interface to avoid disruption [26] – an aspect that can be effectively addressed through a browser extension. At the same time, this and related work highlight concerns regarding the use of AI for CS generation, underlining the importance of preserving users’ authenticity and moral agency in system design [26], [28], [35]. These concerns further emphasize the need to design HAIC systems that support, rather than replace, human decision-making.

With respect to existing HAIC tools, two notable studies stand out. First, Chung *et al.* [32] present a platform designed to support practitioners of Non-Governmental Organizations (NGO) in countering Islamophobic HS on Twitter by offering trend monitoring and suggesting CS based on entries from the CONAN dataset. Their system retrieves pre-written responses using a tf-idf¹ approach rather than generating novel responses with LLMs. Specifically, the input HS instance is compared to entries in the CONAN dataset, and the CS response associated with the most similar HS instance is retrieved and returned as a suggestion without further modification. While their system provides a sophisticated user interface (UI), Chung *et al.* [59] do not leverage AI to generate new CS that is specifically tailored to the given input.

Second, Ding *et al.* [30] introduce *CounterQuill*, an educational platform that emphasizes AI mediation and co-writing rather than full automation. The system integrates learning, brainstorming, and co-writing components aimed at fostering users’ CS-related skills. During the co-writing phase, an OpenAI model generates a draft based on users’ prior

¹Term Frequency–Inverse Document Frequency is a statistical measure used to represent documents in vector space.

reflections, which can then be revised and iteratively refined. Additional features include grammar correction, tone adjustment to enhance empathy, and rewriting based on previously saved brainstorming notes, while users can further tailor outputs by specifying desired tone, structure, or communicative intent. Interview feedback from a case study with 20 participants indicates that this interactive approach supported users in developing their skills, co-writing high-quality CS, and maintaining a sense of authorship. However, as a multi-step process that extends beyond immediate response formulation, the system requires greater user effort and may therefore introduce a higher barrier to engagement. In contrast, the present work aims to provide a lower-threshold solution, while drawing strong conceptual motivation from these findings and their emphasis on supporting effective and authentic CS.

Notably, both described systems require users to switch between platforms for composing and posting responses, introducing additional friction that may hinder practical adoption. The authors of *CounterQuill* themselves acknowledge that, for real-world applicability, such a system should ultimately be implemented as a lightweight browser extension [30]. To the best of current knowledge and at the time of writing, no publicly available browser extension exists that enables users to compose CS directly within their active browsing context. While some extensions can automatically detect and hide or blur HS², none actively support users in generating and contributing CS as a response. This highlights a clear gap that the present work seeks to address.

²e.g., Filter Everything Online, Blockparty, Hate Speech Blocker, WordBlocker

3 System Development

This chapter provides a detailed account of the system design and introduces the framework applied to ensure ethics-by-design. In addition, it outlines the overall architecture of the CS writing assistant.

3.1 Use Case and System Design

To conceptually frame and structure the browser extension, this work is positioned within the design space for writing assistants proposed by M. Lee *et al.* [60]. The authors establish a shared vocabulary and provide researchers with a practical tool to understand different types of writing assistants, as well as support in the conceptualization and design of new systems. They systematically reviewed 115 papers about writing assistants and clustered their findings in five aspects: Task, User, Technology, Interaction, and Ecosystem. Within each aspect, the design space defines dimensions (i.e., principal components of an aspect) and codes (i.e., potential options for each dimension). Using this framework helped define the use case and provided a solid foundation for the development of the CS browser extension, while ensuring alignment with established design principles. The following paragraphs outline the extension’s design with respect to the proposed dimensions and *codes*.

Task

The task dimension characterizes the writing situation in which the assistant operates. With respect to the writing stage, the system supports users in both *drafting* and *revision* of CS, as it provides drafts for given ideas as well as refinement of existing text. Targeting lay users rather than professional NGO workers, the writing context is *personal*, as individuals publicly respond to HS by expressing their thoughts, emotions, and experiences. The audience of the generated CS is clearly *specified* as the author of the HS, its targets, and bystanders within the broader online community. The purpose of CS can vary depending on the situation and the user’s intention, and should accordingly be adapted to being *persuasive*, *analytical*, *educational*, or *feedback-providing*. In the typical use case, task specificity is defined with *general direction*, meaning users provide high-level guidance (e.g., style or length) without specifying exact rhetorical strategies.

User

The browser extension targets users with a broad demographic profile, addressing individuals of diverse backgrounds (e.g., gender, race, age), with the current limitation that the prototype is available only in English. Users' *language and culture* fundamentally shape their writing practices, and perceptions of both HS and CS may vary across cultural contexts. The system does not explicitly integrate detailed cultural knowledge – potential biases are discussed in Section 4.6.

Next, the CS writing assistant aims to enhance several user capabilities. It strengthens *writing expertise* by improving the quality and effectiveness of CS through research-based insights, fostering *empathy* and supporting users in regulating negative *emotions*. It also aims to increase users' *confidence* in both the writing process and the output, resulting in greater willingness to post CS.

As users interact with the extension, they develop a relationship to the system that shapes engagement. A central objective of this work is to create AI-assisted CS that still feels authentic, therefore particular emphasis is placed on fostering *agency* and a sense of *ownership*. Moreover, compared to human support, the extension is characterized by its constant *availability*, offering assistance whenever needed.

Finally, different use cases prioritize distinct system output preferences. In the context of the CS writing assistant, user experience is particularly influenced by *textual diversity* and the degree of *personalization* to an individual's writing style.

Technology

Regarding the underlying model type, the CS writing assistant is built upon a *foundation model*, leveraging a large, pre-trained LLM. To tailor it to CS generation, the system leverages external resource access to exemplar *data* stored in a vector index. A high-quality expert data source is emphasized: the Multitarget CONAN dataset was developed by *experts* with *machine* assistance in an iterative process, resulting in a *medium* data size of about 5,000 labeled examples (see Section 4.3.1).

Depending on user input, the learning problem is framed as either a *rewriting* or a *generation* task. Training and adaption of the model for the task is implemented via *prompt engineering*, including *retrieval* of semantically similar HS/CS pairs to enable ICL. As such, the learning algorithm is based on *supervised learning*.

With respect to scalability, the system design prioritizes limiting *cost* by relying on a pre-trained LLM combined with prompt engineering rather than resource-intensive fine-tuning. *Latency* is optimized through persistent vector storage for retrieval and efficient handling of external API calls.

Finally, following a human-in-the-loop approach, *human evaluation* acts as the primary evaluator. The system is assessed in a user study where participants place their evaluation focus on aspects such as *linguistic quality*, *controllability*, *style & adequacy*, and *ethics*.

Interaction

The interaction between user and writing assistant comprises three central components: the user, the UI (frontend), and the system (backend). M. Lee *et al.* [60] define different interaction metaphors, which shape how the user relates to the system. In the case of the CS writing assistant, it is conceptualized as a *tool*, presenting structured suggestions through conventional UI elements (e.g., buttons, checkboxes) that distribute functionalities across clearly defined controls. The UI is implemented as a pop-up window, representing a layout that clearly separates users' social media activity and AI-generated content, while allowing to stay within the browsing window.

Concerning initiation, the system operates in a fully *user-initiated* manner: CS suggestions are generated only when the user actively triggers the respective button. This design ensures *explicit* control, emphasizing the user's role in steering the system. Outputs are never inserted automatically into the webpage. Instead, users must deliberately integrate system output through *selection* of a suggestion, optionally preceded by iterative *editing*.

The system accesses both free-form *input text* and structured *additional data*, such as the user's role or preferred style and length. Subsequently, the system outputs multiple *generated* CS suggestions, which are clearly visually differentiated from user-written input by being displayed in a distinct *location*.

Ecosystem

M. Lee *et al.* [60] highlight the importance of viewing writing assistants within their sociotechnical environment. While much of the ecosystem may be beyond immediate control, anticipating its influences helps ensure seamless embedding in the existing digital infrastructure. For example, implementing the tool as a Manifest V3 extension ensures *technical interoperability* with diverse browser preferences, making it compatible with Safari, Firefox, and Chromium-based browsers (i.e., Chrome, Edge, and Opera). The extension assumes *local writing* as the primary locale, expecting users to have the necessary technical infrastructure, including a stable internet connection for API access. To account for social factors, the development builds on prior qualitative research exploring users' needs for AI-supported CS writing [26], [28]. Further, a user study is conducted with the prototype to collect feedback from *stakeholders*.

Finally, as a research project contributing to work on combating online HS, the browser extension can be accessed as *free and open-source software*, promoting transparency, reproducibility, and collaboration. The design respects relevant norms and rules, particularly compliance with data privacy *laws*, and reflects on potential impacts on *authors* and how their *writing* may change over time when using the system. Further ethical considerations are detailed in the following section.

3.2 Ethics-by-Design

Since the projects deals with sensitive and potentially harmful content, ethical considerations were explicitly foregrounded in the development of the browser extension through an ethics-by-design approach. Inspired by Kiritchenko *et al.* [14] and Chaudhary *et al.* [61], this process was guided by the eight core principles of ethical AI as identified in a study by the Berkman Klein Center for Internet & Society at Harvard University [62]: privacy, accountability, safety and security, transparency and explainability, fairness and non-discrimination, human control of technology, professional responsibility, and promotion of human values. These principles synthesize recurring ethical themes found across a wide range of AI governance frameworks. Specifically, they were derived from an analysis of 36 influential policy and guideline documents produced by governmental bodies, private-sector actors, professional associations, advocacy organizations, or multi-stakeholder initiatives spanning multiple global regions. All eight principles were thoughtfully studied and considered throughout the system design and deployment process. By adopting a HAIC paradigm, the system inherently addresses key principles such as accountability and human control of technology, thereby establishing a strong ethical foundation that guides its subsequent implementation. A detailed and concrete account of how each of the principles was translated into specific technical and organizational measures is provided in Section 4.6.

3.3 System Architecture

The system follows a modular client–server architecture composed of a frontend, a backend service, a retrieval module, and a LLM accessed through an external API. An overview of the system architecture is shown in Figure 3.1. The frontend, implemented as a browser extension using Manifest V3 along with HTML, CSS, and JavaScript, provides the graphical UI and handles user interaction via a pop-up window. Asynchronous requests are transmitted to the Python-scripted backend via HTTP, where a FastAPI service coordinates the central processing workflow and manages the application’s state. Upon receiving a request, the backend first performs a similarity search within the FAISS vector index to retrieve relevant examples from the Multitarget CONAN dataset, which serve as the foundation for ICL. Subsequently, a structured prompt is constructed – integrating the retrieved context with the user’s input – and forwarded to a LLaMA 3 model hosted via the Groq API. The resulting model output is then returned to the backend, post-processed, and forwarded to the frontend for display. The backend service is deployed on the Railway cloud platform, which pro-

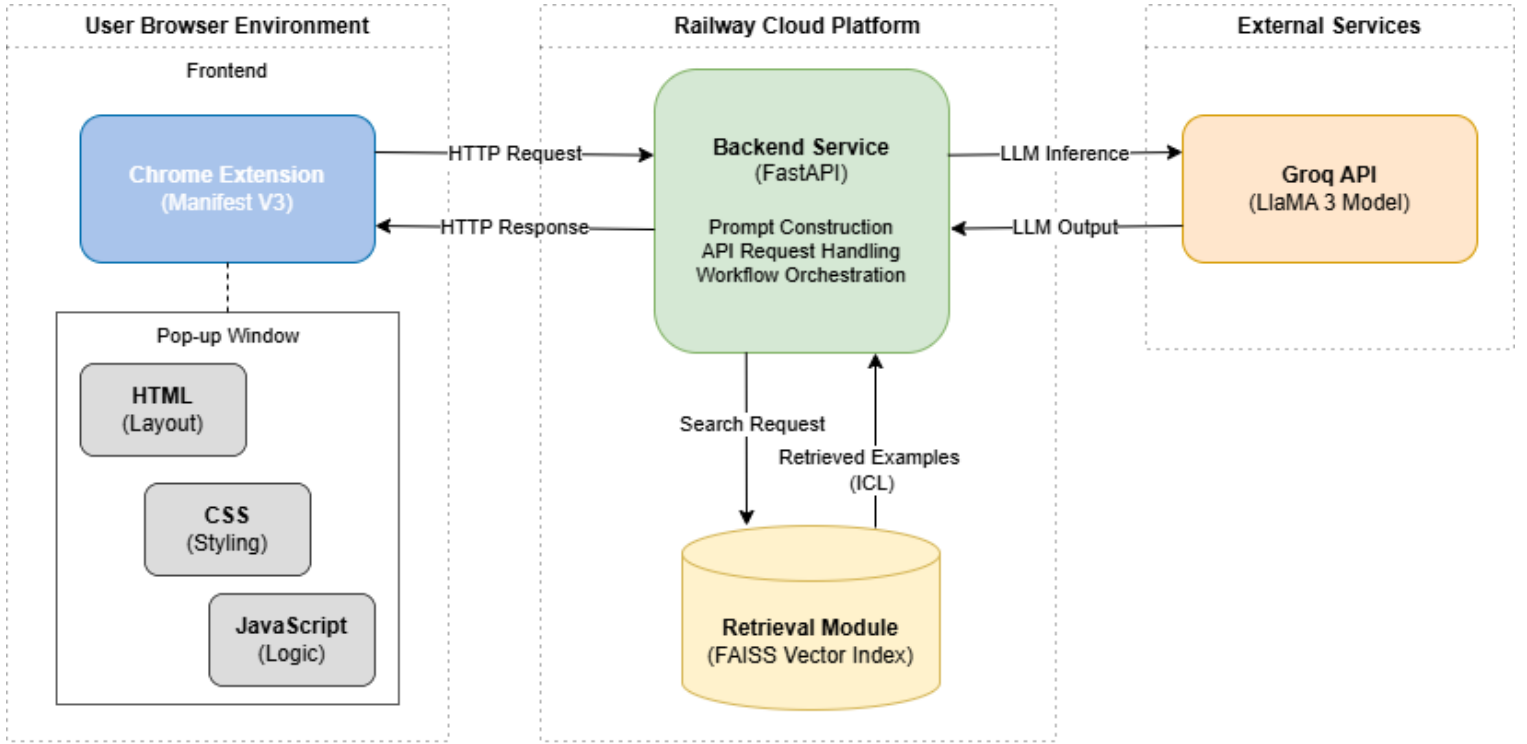


Figure 3.1: System architecture diagram

vides a scalable and robust runtime environment for the API and the retrieval logic. By hosting the FastAPI application remotely, the system eliminates the need for local deployment, thereby removing technical barriers (such as environment configuration or command-line proficiency) that would otherwise be required of users. This modular design effectively separates user interaction, processing logic, data retrieval, and model inference into distinct layers, facilitating independent development and a clear separation of concerns. A comprehensive description of the individual implementation details for each component is provided in Chapter 4.

4 Implementation

This chapter details the concrete implementation of each system component, including the frontend, backend, example retriever, LLM, and deployment. It further outlines how the eight core principles of ethical AI were considered and incorporated throughout the entire system. The GitHub repository containing the full implementation is listed in the Appendix.

4.1 Frontend

The frontend of the system is implemented as a Manifest V3¹ browser extension using HTML, CSS, and JavaScript, providing an accessible and low-threshold UI for the CS writing assistant. It is designed as a compact pop-up window that can be invoked directly within the browser environment, thereby embedding the intervention smoothly into users' online interactions. Within this architecture, HTML structures the content and layout of the pop-up window, CSS defines its visual presentation and styling, and JavaScript enables interactive functionality and communication with the underlying generation process. Visually, the UI follows a consistent two-color scheme (purple-blue and orange) to create an engaging aesthetic while ensuring sufficient contrast and readability. The design prioritizes clarity, intuitive navigation, and transparency of interaction processes.

The pop-up interface is divided into two main views: (1) the Generator window and (2) the Info page. The Generator window constitutes the core functional component. At the top, users are provided with a text field in which they can paste the hateful comment to which they intend to respond. Below this, a mandatory free-form input field labeled "Your Input for the Counterspeech" encourages users to contribute ideas, text snippets or personal experiences. This design decision explicitly aims to promote co-creation rather than passive consumption of automated output, reinforcing user agency right from the start of the generation process.

To further contextualize the generated CS, users must indicate their relationship to the HS (Direct Target, Target Group Member, or Ally) and select a preferred writing style (Familiar or Formal). These structured inputs guide the underlying prompt

¹<https://developer.chrome.com/docs/extensions/develop/migrate/what-is-mv3>

construction and ensure that the generated responses are situationally appropriate and rhetorically aligned with the user’s communicative intent. If the user does not provide input for any of the mandatory fields, an error messages tells them to do so.

In addition, two optional personalization features are provided. A response length slider allows users to determine the approximate length of the generated text, ranging from short (20–40 words) to medium (40–80 words, default setting) and long (80–120 words). This enables adaptation to different platform norms and attention spans. Furthermore, users may activate a checkbox labeled “Use Placeholders for personal details.” When enabled, the system incorporates placeholder segments in square brackets into the suggestions, which the user can later replace with personal anecdotes or details. This feature explicitly encourages users to post-edit the generated output. As both the placeholder function and the role selection may not be immediately self-explanatory, each is accompanied by an adjacent information icon providing definitions.

Upon clicking the “Generate responses” button, an asynchronous HTTP request is sent to the backend API. During processing, a status message is displayed to provide immediate feedback and prevent perceived latency. Once the backend returns an HTTP response containing three CS suggestions, they are displayed in separate text boxes. By presenting multiple alternatives rather than a single output, the system encourages critical evaluation, reflective engagement and user choice. Each suggestion is accompanied by two action buttons: “Edit”, which allows direct modification within the field, and “Copy”, which facilitates easy transfer to the target platform (or iteration within the extension). Overall, the Generation window is designed to optimize user experience while actively fostering and encouraging user engagement. Screenshots of the Generator page can be found in Figures A.1, A.2 and A.3.

The second window, the Info page, addresses ethical and legal considerations to ensure transparency and informed participation. It presents three informational sections covering Data Privacy, Accountability, and Ethical Use, which inform about data handling and user responsibility when using the writing assistant. Users must actively acknowledge these conditions by enabling a consent toggle, otherwise the generator will throw a respective error button. Importantly, consent can be withdrawn at any time by disabling or uninstalling the extension. The three consent statements can be found in Table A.1.

4.2 Backend

The backend of the application is implemented in Python using FastAPI², which handles API requests and responses, and coordinates model interaction. LangChain³ is employed as an integration framework to integrate all components into a coherent backend pipeline. More specifically, LangChain simplifies the orchestration of LLM calls and enables a seamless incorporation of the external vector store used for example retrieval.

The backend exposes a single REST⁴ endpoint `/generate`, which receives all user-submitted input for CS generation. Upon receiving a request, the endpoint validates the input parameters, retrieves semantically similar examples from the retriever, constructs the final prompt, and submits it to the external Groq API that hosts the LLM. The returned model output is then parsed into structured suggestions, and sent back to the frontend for display. As the system's prototype is developed for controlled experimental use, no user-level authentication mechanism is implemented for the endpoint. Potential errors during API calls, such as timeouts or invalid responses, are handled by returning structured error messages to the frontend in order to prevent application crashes.

Prompt construction follows the structure required by the external model host provider Groq, which distinguishes between a system message and prompt content. The system message defines the overall task instruction, relevant guidelines including the specified effectiveness quality criteria, and safeguards. On the other hand, the prompt content contains all user-provided input, specifies the desired output format, and provides step-by-step instructions for generating CS. It further establishes a priority order to ensure goal-aligned and controlled generation (Safeguards > Default guidelines > User input > Retrieved examples). The templates for the system message and prompt content are provided in A.3 and A.4, respectively.

4.3 Example Retriever

As part of the prompt engineering, HS/CS examples are provided to the model for demonstration. Such ICL is expected to enhance the quality of generated responses while avoiding the computational cost of fine-tuning (see Section 2.2).

²<https://fastapi.tiangolo.com/>

³<https://www.langchain.com/>

⁴<https://restfulapi.net/>

4.3.1 Data

For providing examples to prompts for ICL, the Multitarget CONAN dataset released by Fanton *et al.* [63] was deemed most appropriate because it is the only available dataset that provides high quantity, diversity of targets and the CS quality of interest. It features 5003 HS/CS pairs in the English language, labeled with eight common targets of hate: DISABLED, JEWS, LGBT+, MIGRANTS, MUSLIMS, POC, WOMEN, OTHER. The dataset was collected via a hybrid approach of automatic generation and niche sourcing. Specifically, HS/CS pair proposals generated by GPT-2 are reviewed and post-edited by three expert NGO operators. Multitarget CONAN appears to be the current gold dataset and is used in several studies for fine-tuning, augmenting or evaluating LLMs [49], [51], [52], [64]. Other existing datasets are less suitable for the goal of this study because they either focus on a single target group [59], only include independent HS and CS labels for binary classification [39], [64], the CS was written by non-experts [65]–[68] or fully by generative models without human intervention [6], [69].

4.3.2 Embedding and Retrieval

Upon initialization, the implemented retrieval service loads the dataset from a CSV file containing the HS/CS pairs. Each HS instance is then embedded using the model *all-mpnet-base-v2*⁵ [70], which is the best-performing model in the Sentence-Transformers library and has demonstrated strong capabilities for semantic similarity tasks⁶. The resulting dense representations are stored in a FAISS⁷ vector index, which provides efficient and reliable semantic similarity search over high-dimensional sentence embeddings. To improve runtime performance, the index is persisted locally and reloaded on subsequent starts if available.

When a user submits a CS generation request, the given HS instance is embedded with the same embedding model and compared against the indexed examples using k-nearest-neighbor search. The retriever then returns the top-5 most semantically similar HS instances from the Multitarget CONAN dataset along with their associated CS. These five HS/CS pairs are then dynamically integrated into the model prompt for ICL.

⁵<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

⁶https://sbert.net/docs/sentence_transformer/pretrained_models.html

⁷<https://faiss.ai/index.html>

4.4 Large Language Model Integration

The backbone of the writing assistant is a LLM, namely LLaMA 3.3 70B, which is a 70-billion-parameter model from the LLaMA 3 family [71]. This choice was made due to LLaMA’s strong language capabilities and open-source transparency [72]. Its development followed the paradigm that optimal performance is not necessarily achieved through the largest possible models (i.e., with many parameters and tokens), but rather through compute-efficient architectures trained on large amounts of data [73]. Although such extensive training increases carbon emissions upfront, smaller models require less energy and GPU resources during inference, making them more efficient in practical deployment. Additionally, unlike many other models, including GPT-3, LLaMA models were trained exclusively on publicly available data, allowing for high transparency and supporting the objective of open and reproducible research [71], [72].

The model is accessed via the Groq API⁸, which provides free and efficient API access to state-of-the-art LLMs, enabling high-speed inference without the need to run large models locally or on dedicated deployment servers. This avoids the requirement for powerful local hardware or costly cloud infrastructure. Among the LLaMA variants available through Groq (as of December 2025), the strongest model `llama-3.3-70b-versatile` was chosen to ensure high output quality [71]. As this thesis investigates ICL, the model is used in its default configuration without fine-tuning. It is invoked with a temperature of 1.0 – whereas some CS studies favor lower temperatures for deterministic outputs [57], [58], this work follows approaches that prioritize diversity and creativity in generated responses [49], [54]. Output length is controlled implicitly through prompt instructions rather than token limits or post-processing.

4.5 Deployment

The backend and retrieval service is deployed on Railway⁹ to provide users with easy access to the system without requiring local deployment of the FastAPI application. Railway hosts the backend as a web service and manages the runtime environment, including dependency installation and process management. To avoid hard-coding of sensitive credentials (most importantly the Groq API key) in the repository, they are stored securely using environment variables provided by the deployment platform. With regard to latency, using the Groq API for inference and a persisted vector store for the retriever ensures that CS suggestions are generated within a few seconds, balancing responsiveness with the complexity of the 70B-parameter LLaMA model.

⁸<https://console.groq.com/home>

⁹<https://railway.com/>

As the prototype is primarily intended for experimental use with a limited number of concurrent users, no dedicated horizontal scaling mechanisms are implemented. However, the cloud-based hosting setup allows for resource adjustments if higher traffic or additional computational demands arise in future versions.

4.6 Ethical Considerations in Implementation

The eight core principles of ethical AI published by Fjeld *et al.* [62] are adapted in this work to the specific context of a browser extension that embeds AI within a broader user-facing workflow. As such, the ethical assessment and implementation does not focus exclusively on the underlying LLM, but also considers interaction design, user agency, deployment context, and potential misuse. Known limitations and trade-offs are explicitly acknowledged to ensure a transparent and reflective application of the ethical framework.

Privacy

The principle of privacy emphasizes that AI systems should limit data use and ensure that users retain control over how their personal information are handled [62]. Such privacy considerations are addressed in accordance with the General Data Protection Regulation of the European Union (GDPR)¹⁰, particularly focusing on data minimization and informed consent. In concrete terms, the browser extension does not scan or monitor the Document Object Model (DOM) of visited websites but only processes individual comments upon explicit user request. No identifying information is required for use of the extension, and no additional user data are collected beyond the content explicitly entered by the user. Since user-provided content may contain sensitive personal data (such as political opinions or references to sexual orientation) users are informed and asked to consent to the transfer of input data to the LLM. Consent can be withdrawn at any time by disabling or uninstalling the extension. However, users are informed that data previously transmitted to the LLM cannot be retroactively deleted. In terms of interaction data, the prototype does not employ tracking mechanisms or analytics tools and does not make use of persistent identifiers (such as cookies).

Concerning the Railway deployment, a server hosted in Amsterdam, Netherlands, was chosen to ensure that core application infrastructure remains within the European Union. However, because the application relies on the external Groq API, there is limited control over the geographic location of downstream processing, and it is therefore possible that data transmitted to Groq may be processed outside the EU.

¹⁰<https://gdpr-info.eu/>

Finally, the privacy-first approach extends to the source material: since the Multitarget CONAN dataset was authored and reviewed by experts rather than being collected from online conversations, it eliminates any risk to individual rights or personal privacy of online users.

Accountability

Accountability principles emphasize the need for clear mechanisms that assign responsibility for the impacts of AI systems and provide remedies when harms occur [62]. A key limitation of the extension is that the underlying LLM is provided by an external service, leaving aspects such as training data, internal processes, and output behavior beyond direct control. To address this, accountability is ensured through explicit human oversight, with users retaining full responsibility for any content they publish. This is made explicit during installation, where users acknowledge and consent their accountability for all actions and resulting consequences. By clearly assigning decision-making authority to the user, the system reinforces human agency and avoids delegating moral responsibility to the AI.

Safety and Security

The principles of safety and security require that AI systems operate reliably as intended while remaining robust against misuse, failure, and unauthorized interference [62]. The browser extension addresses these requirements through a combination of technical and procedural safeguards. Sensitive credentials (e.g., the Groq API key) are securely stored as environment variables within the hosting platform and are never exposed in public repositories. All communication with external services is encrypted via HTTPS, ensuring the protection of data in transit.

At the same time, the persuasive nature of CS introduces a risk of misuse, as the system could potentially be repurposed to reinforce harmful viewpoints rather than counter them [74]. To mitigate this risk, safeguards are implemented at the prompt level. Drawing on the guardrail approach proposed by Ding *et al.* [30]¹¹, the LLM is instructed to evaluate whether user-provided input contains harmful intent and, if so, to reject its incorporation and generate only non-toxic responses.

Transparency and Explainability

Transparency and explainability principles call for AI systems to be designed in ways that enable meaningful oversight, including clear communication about system operation, usage contexts, and limitations [62]. Transparency is addressed by clearly informing users after installation about how the extension operates, including which

¹¹See their GitHub repository for details: <https://github.com/xding2/CounterQuill>

AI model is used and where user-provided data are transmitted to for processing. The source code of the system is publicly available in Github (see Appendix).

The model choice further supports transparency: LLaMA 3 is an open-source LLM for which architectural details and high-level information about training data are publicly documented, which would not have been the case for most proprietary alternatives [71]. Because the extension functions as an assistive tool and does not make autonomous decisions, explainability of individual outcomes is of limited applicability.

Fairness and Non-discrimination

The principle of fairness and non-discrimination requires AI systems to be designed and deployed to mitigate bias, promote inclusivity, and avoid disadvantaging individuals or groups [62]. The browser extension supports this by not collecting demographic user data (unless explicitly provided), thereby avoiding differential treatment based on user characteristics. In addition, the safeguards described in the *Safety and Security* section aim to prevent the generation of toxic or discriminatory content targeting individuals or groups.

However, adherence to this principle is inherently limited by the underlying AI model. Prior research shows that LLMs, including LLaMA, can reproduce social biases and stereotypes present in their training data [72], [75]–[78]. To address this, one of the consent statements explicitly informs users that generated outputs may reflect such biases and should be critically evaluated.

A further limitation stems from the nature of the task itself: processing abusive input is necessary for CS generation, but such exposure may reinforce the model’s capacity to reproduce harmful language.

Human Control of Technology

The principle of human control of technology requires that consequential decisions involving AI systems remain subject to human judgment [62]. While the prompt design of the underlying model aims to minimize the risk of toxic or biased outputs, even advanced LLMs may still produce undesirable content. Addressing this limitation is central to the tool’s design as a human-in-the-loop system: all generated CS suggestions are presented for user review, allowing validation, modification, and post-editing before any use or publication. Although users cannot control the initial generation, they retain full authority over the final output, ensuring that decision-making and responsibility remain with the human rather than the system.

Professional Responsibility

The principle of professional responsibility emphasizes that developers must anticipate the societal impacts of AI systems and act with integrity throughout design and

deployment [62]. In the context of CS generation, a key concern is that AI-assisted responses may become overly generic, diminishing users’ personal voice both in individual interactions and, cumulatively, in online discourse [26]. Preserving authenticity is therefore a central design objective through active engagement of users in the CS generation process.

At the same time, professional responsibility entails limiting unintended influence on users. Given that perceptions of toxicity vary and automated interventions may introduce bias [74], the system deliberately avoids automatic HS detection or nudging users to respond. Instead, users retain full autonomy in deciding whether and when to engage, with suggestions generated only upon explicit request.

Moreover, prior research shows that AI-generated persuasive content can shape not only how users write, but also how they think [79]. When certain viewpoints are systematically favored, LLMs may exert latent persuasion¹² by making some expressions easier to produce than others. To mitigate this risk, the system promotes active user involvement by facilitating the inclusion of personal input, encouraging users to draw on their own experiences, and providing multiple CS suggestions that reflect diverse perspectives rather than a single predefined stance.

Promotion of Human Values

The principle of promoting human values requires that AI systems are aligned with fundamental societal norms and contribute to human well-being [79]. By design, the browser extension adopts CS as an alternative to interventions such as content removal or censorship, which may conflict with freedom of speech. Instead, it supports users in responding to HS through empathetic, non-toxic communication, with the aim of fostering constructive dialogue. While safeguards in the prompt aim to reduce the likelihood of toxic outputs, occasional failures remain possible, reflecting the broader and still unresolved challenge of reliably detoxifying LLMs [80].

In addition, prior evaluations of LLaMA models highlight their susceptibility to generating misinformation, including hallucinated content [72]. To mitigate this risk, the presented prototype deliberately avoids fact-based argumentation: the prompting strategy explicitly instructs the model not to introduce factual claims or external information, thereby trading expressive power for greater reliability.

Overall, the system is designed to promote key human values such as freedom of expression, respect, empathy, truthfulness, and responsible participation in online discourse.

¹²Unlike explicit persuasion, this influence is often unintentional and difficult to detect, as model preferences may be opaque and context-dependent [79].

5 Experiments and Results

This chapter presents the experiments conducted to assess and evaluate the developed browser extension across the four dimensions of interest: seamless integration into online workflows, perceived authenticity and effectiveness of the generated output, and impact on time requirements. The evaluation design and results for each dimension are described in detail in separate sections. The findings provide insights into the system’s strengths and highlight areas for further refinement.

5.1 Experimental Setup

To answer the research question, the browser extension was tested and evaluated in a user study. This section outlines the experimental study design, gives details on the cohort of participants, and shares ethical considerations guiding the study.

5.1.1 Study Design

The user study followed a two-part design. In the first part, participants completed two rounds in a within-subjects setup. They were presented with eight HS instances from the Multitarget CONAN dataset in a controlled environment to reduce distractions and unpredictability associated with real social media platforms. Instances were semi-randomly selected (with one instance per target group) and curated to ensure suitability and variation in hate intensity, reflecting the diversity of online HS while avoiding overly extreme or generic cases. All participants received the same set (displayed in A.2), which was excluded from the FAISS index to prevent identity retrieval and evaluation bias during ICL – ensuring the model could not fetch the test instance itself as an exemplar¹. All tasks were conducted using Google Chrome.

In the first round, participants selected three out of the eight HS instances (allowing some flexibility based on personal experience or interest) and wrote CS responses without assistance. After each response, they rated their willingness to post the comment, which was the only applicable authenticity measure in the unassisted condition. In

¹To ensure consistency, only exact duplicates were removed. Closely similar instances, like variants differing by only a single sentence, may still remain in the example data.

the second round, participants responded to the same three instances again, this time using the browser extension. Using identical instances across both rounds enabled a direct comparison of willingness to post and effectiveness. After each response, participants evaluated perceived authenticity towards it. Writing time was recorded in both rounds. Following the task, participants completed a questionnaire on usability and user experience. During the assisted round, they shared their screen and were encouraged to think aloud, providing insights into interaction behavior and reasoning behind their ratings.

To complement the quantitative data, participants provided qualitative feedback through a short semi-structured interview (5–10 minutes) at the end of the user study. They were asked to describe their overall experience with the system, reflect on individual features, and suggest potential improvements. Responses were analyzed exploratively by identifying and grouping key themes. While Likert-scale ratings indicate *how* the system was evaluated, qualitative feedback explains *why*, enabling a more comprehensive understanding. Relevant insights are integrated throughout the different results subsections, with minor edits to quotes (i.e. removing of filler words or correcting grammatical inconsistencies) for improved readability while strictly preserving meaning.

The second part of the study was conducted one week later as a survey. Participants peer-evaluated CS written by others in terms of effectiveness, avoiding the need for extra annotators. While defining evaluation criteria requires expertise, rating responses does not and is commonly performed by non-experts [6], [51], [56], [81]. To ensure objectivity, no self-evaluation was conducted. Both the authenticity and effectiveness evaluation capture perceived output quality rather than model performance, as the focus lies on human–AI interaction. Accordingly, only final, potentially post-edited responses were evaluated.

5.1.2 Participants

The participant cohort for the user study comprised five individuals, consisting of three women and two men, with ages ranging from 26 to 32 (mean age = 28.9). The group represented diverse nationalities, including German, Russian, and Uzbek; notably, all participants were proficient in English, utilizing it as either their primary or daily language. Reflecting the common targets of online hostility, the cohort was characterized by significant demographic diversity: two participants identified as members of the LGBTQ+ community, three possessed a migration background, one was Muslim, and three identified as female. In three instances, participants occupied intersectional identities, belonging to multiple marginalized groups, while one participant did not

belong to any of the specified target demographics. None of the participants identified as members of the following minority groups: Disabled, Jewish, or People of Color. Regarding digital habits, all were active across at least three different social media platform, including Instagram, Facebook, Telegram, WhatsApp, and YouTube, among others. However, engagement with CS was low: only one participant had previously authored CS on a few occasions, while the others had no prior experience. Furthermore, while two participants had a professional research background in online hate mitigation, the remainder had no prior specialized knowledge of HS or CS. Participation in the study was entirely voluntary and conducted without financial compensation.

5.1.3 Ethical considerations

To ensure the psychological well-being of participants during the user study, several ethical measures were implemented to mitigate the impact of exposure to offensive or disturbing content. Prior to engagement, all participants – who were adult volunteers fully briefed on the study’s objectives – provided formal informed consent. They were explicitly warned about the content they would encounter, including the specific groups targeted by the HS, both during the initial recruitment and at the start of each study phase. It was clearly communicated that they could withdraw from the study at any time, without giving a reason and without negative consequences. Furthermore, the consent form provided participants with direct contact information for support should they experience any distress or harm as a result of their participation.

5.2 Seamless Integration

This part of the experiments aims to assess how seamlessly the browser extension supports users in composing CS, with a focus on the interaction process and user experience, rather than the quality of the final generated outputs.

5.2.1 Evaluation Design

Seamless integration of the tool was assessed using the User Experience Questionnaire (UEQ)², which was introduced in 2008 by Laugwitz *et al.* [82]. It has since become a well-established instrument for measuring and comparing both soft (user experience) and hard (usability) qualities of interactive digital products, including browser extensions [83], [84] and CS applications [31]. Through an iterative development process, 26 key

²<https://www.ueq-online.org/>

items were identified and grouped into six scales: attractiveness (overall impression and likability), perspicuity (ease of familiarization), dependability (perceived control and predictability), efficiency (effort required to complete tasks), novelty (perceived innovation and originality), and stimulation (degree of motivation and engagement) [85]. The attractiveness scale stands as an independent dimension, while the remaining scales can be further grouped into pragmatic (perspicuity, efficiency, and dependability) and hedonic (stimulation and novelty) quality dimensions [86]. Each item is presented as a pair of opposite adjectives (e.g., boring/exciting). To reduce central tendency bias commonly associated with 5-point Likert scales, the UEQ employs a 7-point scale, ranging from -3 to 3 [85].

In 2017, M. Schrepp *et al.* [87] introduced a short version of the UEQ (UEQ-S), consisting of a reduced set of eight items. These items comprise four pragmatic and four hedonic measures, selected via a principal component analysis of the original dimensions and verified for relevant cross-loadings. As the questionnaire served as the final component of an extensive experimental design, using the UEQ-S (i.e., reducing the number of items from 26 to 8) effectively managed participants' cognitive load, thereby ensuring the quality and reliability of the collected data.

To contextualize the results, they are compared against the official UEQ-S benchmark (as of February 2026). It is derived from the original UEQ benchmark dataset [88], [89], which is based on evaluations from 21,175 users across 468 interactive products. Their scale means are utilized to calculate the benchmark for the short version, as the UEQ-S does not yet possess a sufficiently large independent dataset [88]. Accordingly, the authors note that the results of the calculated UEQ-S benchmark are to be interpreted with care.

The benchmark classifies products into five categories: (1) excellent (top 10%), (2) good (better than 75% of benchmark results), (3) above average (better than 50%), (4) below average (better than only 25%), and (5) bad (among the worst 25%) [89]. This enables a relative positioning of the prototype within a broader ecosystem of digital products. Since the UEQ-S is primarily designed for large-sample quantitative studies, it is acknowledged that the participant number in this project is not sufficient for robust statistical generalization. Nevertheless, the questionnaire is still highly valuable as it provides a structured, standardized, and comparable framework to assess user experience and usability, and helps identify areas for iterative improvements of the early-stage prototype [86].

5.2.2 Results

The UEQ-S is accompanied by a dedicated Excel-based analysis tool that facilitates data interpretation and visualization. Upon analysis of the raw data, no negative ratings

Item pair	Mean	Std. Dev.
obstructive/supportive	2.2	1.1
complicated/easy	2.0	1.0
inefficient/efficient	2.0	1.2
confusing/clear	2.2	0.8
Pragmatic Quality	2.1	0.6
boring/exciting	2.4	0.5
not interesting/interesting	2.6	0.9
conventional/inventive	2.0	1.2
usual/leading edge	1.4	1.1
Hedonic Quality	2.1	0.6
Overall	2.1	0.6

Table 5.1: Mean ratings and standard deviation (Std. Dev.) for each of the eight UEQ-S item pairs, as well as the aggregated pragmatic, hedonic and overall score. Metrics are based on a 7-point Likert scale ranging from -3 (most negative) to 3 (most positive)

were observed, and all evaluations met or exceeded the neutral threshold zero. Table 5.1 shows mean scores and standard deviations for each individual item pair of the UEQ-S, as well as aggregated for the pragmatic and hedonic dimensions, and the overall global score. However, given the small number of participants, these statistical outputs should be interpreted with caution. According to the provided UEQ-S sample size analysis for estimating scale means, a minimum of seven participants would have been required to achieve a precision of 0.5 with an error probability of 0.1. Consequently, the following values are intended to provide an initial high-level comprehension of how participants evaluated the browser extension.

The results show that participants perceived the extension as particularly *interesting* and *exciting*, with these items yielding high mean values (2.6 and 2.4, respectively) and standard deviations below 1. The lowest-scoring item pair was *leading edge*, which achieved a mean of 1.4, which is still in the middle of the positive spectrum. Both the pragmatic and hedonic quality dimensions yielded an identical average scale of 2.1 (resulting in a global mean of 2.1 across all items). Notably, the individual mean values for the four hedonic items exhibited greater variance than those of the pragmatic ones. This suggests that while participants held a highly consistent perception of the

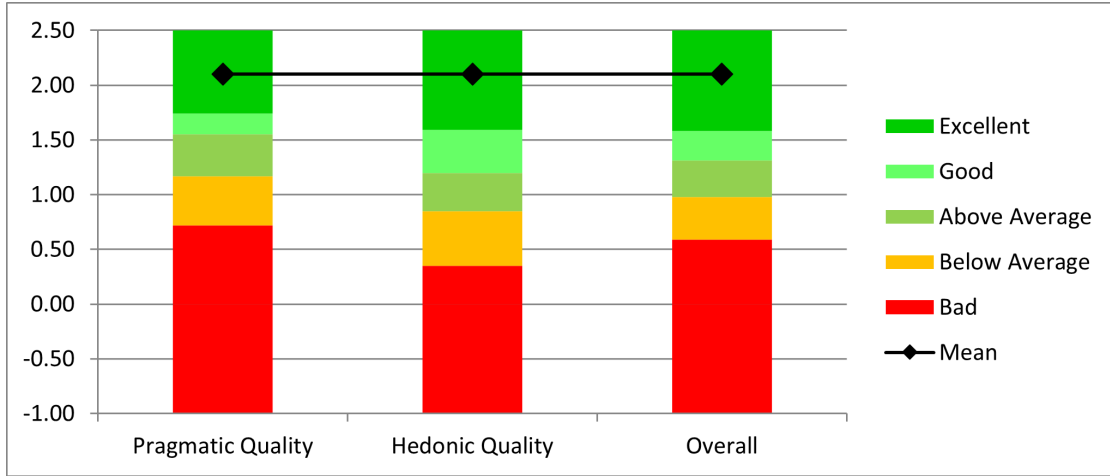


Figure 5.1: Comparison of mean ratings against the UEQ-S benchmark

tool’s task-related utility (in terms of perspicuity, efficiency, and dependability), their subjective experiences regarding the extension’s novelty and stimulation were more diverse.

According to M. Schrepp *et al.* [89], mean values exceeding +2 or −2 are rare, as respondents frequently avoid extreme answer categories. Correspondingly, when measured against the established benchmark, the browser extension consistently falls into the "excellent" category, as illustrated in Figure 5.1³. The black data points represent the ratings from the user study across pragmatic and hedonic qualities, as well as the overall score, while the color-coded scale indicates the percentile ranks of the benchmark. Notably, the tool performed on par with the top 10% of all products represented in the benchmark dataset.

The interview feedback reinforced the overall positive ratings, with participants expressing an enthusiastic overall impression about the intuitive, well-structured UI and aesthetically pleasing visuals. They felt the feature set was well-balanced, offering various personalization options without becoming overwhelming, and noted that the LLM’s processing time of a few seconds was acceptable. One participant shared that facing abusive comments often left them "*speechless*", having a strong opinion but not knowing how to put it into words. They highly valued the extension’s ability to help overcome this mental block, providing the right language to turn their thoughts into a clear and convincing response. Two participants specifically reported that the tool was

³To maintain methodological consistency and comparative accuracy, the visualization from the UEQ-S analysis tool is retained, accounting for the stylistic deviation from other figures in this work.

very supportive in summarizing, reordering and structuring their arguments. Another one appreciated that the tool provided a pre-written prompt framework for the specific context:

"I think it's a very useful tool, especially within the browser. Maybe other people would go over to ChatGPT or some other large language model and would look for something there. So you already have a fairly controlled area that is tailored to it. I think that's pretty good."

Constructive feedback was also expressed. A participant with visual impairments noted that larger fonts and higher contrast would improve readability and foster more inclusive design. A recurring functional pain point was the lack of data persistence. Users found it frustrating when accidentally clicking away the window caused their progress to vanish, revealing a clear need for input caching. Similarly, a participant wished for a notebook feature to save and compare different generated drafts for later selection. Another suggested to save default settings or create a profile that may even learn the user's persona and eventually fill placeholders automatically with personal details.

Furthermore, there was missing clarity regarding the input field. Participants used the space differently, and satisfaction was notably higher when they provided a text draft rather than abstract instructions. Additionally, one noted that the mandatory input field could become *"annoying"* during regular use, despite acknowledging that it improves response quality. Viewing the extension primarily as a time-saving tool, they suggested that user input should be optional or integrated into a flexible brainstorming step to better support rapid drafting.

The role selection feature was universally appreciated, as participants noted that being personally affected by HS *"makes a big difference in [...] how sensitive you may formulate the answer"*. However, there was some lack of clarity regarding exactly how role selection influenced the final output. It should further be noted that the "Direct Target" role could not be specifically tested within this user study. Regarding the length specification, one participant found the three-option slider unintuitive, suggesting separate buttons (consistent with the role selection) as a better format. While the short and medium length options were popular, the long setting was rarely used, as participants felt it exceeded the typical attention span of social media or lacked substantive depth.

Beyond existing features, two participants expressed a desire for a more co-drafting oriented design to support iterative refinement, allowing them to automatically copy suggestions into the input field, provide feedback on specific phrases or mark sub-sections of a response for regeneration. To further facilitate the workflow, a participant suggested moving beyond manual copy-pasting toward direct DOM interaction, such as a pop-up button appearing next to social media comment fields to trigger the assistant instantly. Lastly, one participant expressed their wish for more factual support.

Apart from explicit feedback, usage patterns further revealed that features like the placeholders and editing option were often overlooked due to their visual hierarchy and non-intuitive use. For instance, the "edit" button's white color inadvertently nudged users toward the blue "copy" button, while the hover-only info icons were frequently mistaken for broken buttons when trying to click on them. On the info page, only one participant actually read all the consent statements before activating the consent toggle, which revealed a need for more engaging ways to ensure users understand their responsibilities.

Despite initial enthusiasm, some participants said they might still refrain from using the extension due to a skepticism that CS is inherently pointless or ineffective. To address this, one participant proposed a dedicated educational section to define the goals of CS and clarify effective strategies. Beyond the drafting process, they also suggested including resources such as helplines and links to external initiatives for those affected by HS, positioning the extension as a broader gateway for support and advocacy.

5.3 Authenticity

The authenticity evaluation aimed to capture the user's relationship to the generated CS, thus whether it feels authentic to them. Since most AI-generated CS research focuses on fully automated systems without human involvement, there is little research in this area and no existing benchmark for evaluating how users feel about AI-assisted CS writing. Existing work on HAIC in CS generation primarily refers to authenticity and agency without providing adequate definition [26], [28], [30], [35]. Furthermore, all of these studies use qualitative interviews to capture such dimensions, but the current research lacks a standardized terminology and comparable framework. This study attempts to fill this gap by proposing five criteria, which can be measured on a Likert scale. "Authentic"⁴ is used as an umbrella term to assess whether users feel a sense of ownership over the co-authored text, whether it feels personal in both language and content, and whether they would feel comfortable posting it under their own name. Future research may challenge and adapt the proposed terminology and framework.

5.3.1 Evaluation Design

In an initial attempt, sense of ownership was intended as an authenticity criterion [28], [30]. However, the concept is broad, multifaceted, and inconsistently defined in the liter-

⁴Based on the second Merriam-Webster Dictionary definition, "true to one's own personality, spirit, or character" (<https://www.merriam-webster.com/dictionary/authentic>) and the Stanford Encyclopedia of Philosophy's description, "of undisputed origin or authorship" or a "reliable, accurate representation" (<https://plato.stanford.edu/entries/authenticity/>).

ature. For example, the terms "authorship" and "ownership" are frequently conflated or used interchangeably [90]. Despite their conceptual relatedness, they express different ideas. Authorship focuses on whether users feel that they wrote the text themselves [91]. Psychological ownership, on the other hand, is a concept introduced by Pierce *et al.* [92] and describes a cognitive–affective attitude that includes both intellectual acknowledgment and emotional attachment. Subsequent work has further researched the concept and adapted it for digital contexts [93]–[96]. In a comprehensive review, Kuzminykh *et al.* [94] identified five relevant dimensions of psychological ownership in HAIC: self-identity, self-efficacy, accountability and responsibility, autonomy, and territoriality. To avoid overloading the ownership axis, these dimensions were translated and condensed into three criteria: self-identity, sense of agency, and accountability. Two additional dimensions, naturalness and willingness to post, were added to capture the authentic character of CS. To provide explanations and reasoning for these criteria, they are described in detail below.

Self-identity

Self-identity concerns the extent to which an object is perceived as an expression, extension, or representation of the self [92], [94]. This dimension assesses whether users feel the CS reflects their thoughts, feelings, and values. The emphasis lies on alignment with content rather than linguistic style.

Sense of Agency

Mieczkowski *et al.* [97] describe agency in HAIC through two recurring components: control and action initiation. Accordingly, this term combines self-efficacy (i.e., users' perceived capability to perform a task and influence the outcome) and autonomy (i.e., their ability to initiate decisions and actions) [94]. Sense of agency refers to the corresponding conscious experience of these capacities [98]. It focuses on the CS generation process rather than the final output, measuring whether users feel in control and actively engaged throughout.

Accountability

The remaining ownership-related dimensions identified by Kuzminykh *et al.* [94] are accountability and responsibility, as well as territoriality. The first two describe the voluntary or enforced obligation to take care of an item and accept its consequences, whereas territoriality refers to a sense of possession and the readiness to defend it if challenged [94]. These aspects were combined into a single criterion called accountability, which assesses whether users stand behind the generated CS and would accept responsibility for potential outcomes rather than blaming it on the AI.

Naturalness

Prior CS research often evaluates linguistic features in terms of language quality [53], correctness [99], grammaticality [48], or fluency [57], [100], [101]. As modern LLMs generally produce coherent and grammatically sound text [6], such metrics were omitted from the effectiveness evaluation. Instead, following H. Wang *et al.* [42], naturalness captures whether the CS reads smoothly, is easy to understand, and resembles how a human would naturally write. It is included in the authenticity evaluation because it is measured from a subjective point of view, incorporating the idea of personalization and extending fluency to personal style [90]. It is assumed that a CS feels much more authentic to a user if style, tone and expression mirror their language habits and sound like them.

Willingness to Post

Jiang *et al.* [81] and Tekiroğlu *et al.* [48] introduced a binary "choose-or-not" metric to assess whether users would select a CS for real-world use (potentially after post-editing). However, "choosing" does not necessarily imply the action of actually posting it themselves, and the binary format lacks nuance. To better capture how comfortable users would feel posting a CS publicly under their own identity, this study introduces willingness to post. It encourages participants to imagine a real-world scenario rather than an anonymous experimental setting, allowing for a more genuine evaluation of personal comfort.

Participants rated each generated CS instance individually across the five dimensions using a 5-point Likert scale, to align with the effectiveness evaluation (see 5.4.1). The exact wording of the questions as presented to the participants can be found in Table 5.2.

5.3.2 Results

Willingness to post represented the only authenticity criterion evaluated in the first round, because all other metrics were inherent to self-authored text. Figure 5.2 illustrates the average Likert scale ratings of willingness to post for each participant across both rounds, with individual participants represented by distinct colors. Each data point signifies the participant's mean rating across the three CS responses within a round. Since the dotted data trajectories remain vertically isolated (showing no overlap, with one exception), it is evident that participants held unique baseline comfort levels for posting under their own identity. In two instances, the presence of writing assistance had no measurable effect on the willingness to post, suggesting a stable level of confidence in online self-expression. Conversely, for the remaining three participants,

Criteria	Definition
Self-identity	The CS reflects my thoughts, feelings, and values.
Sense of Agency	I felt actively engaged in the writing process of the CS.
Accountability	I stand behind the generated CS and feel accountable for it.
Naturalness	The style and tone of the CS reflect how I would naturally write.
Willingness to Post	I would feel comfortable posting this CS publicly under my own identity.

Table 5.2: Authenticity evaluation criteria and their definition as presented to participants

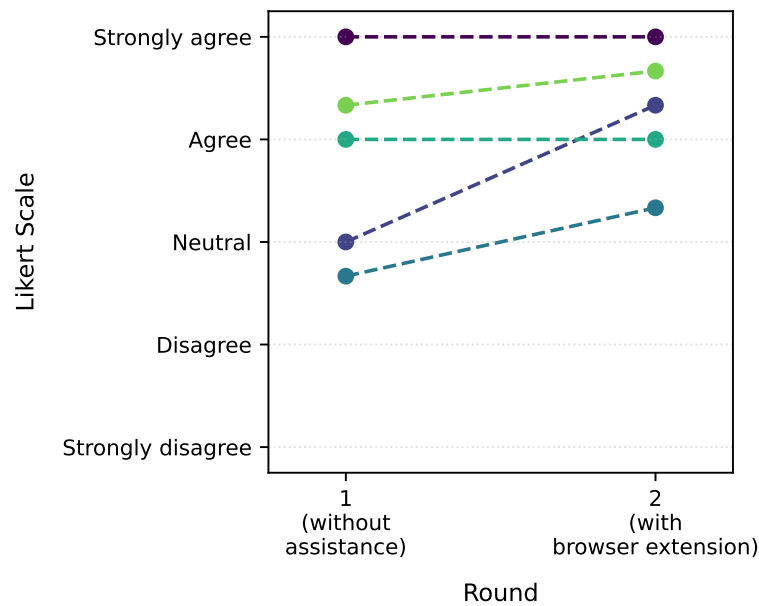


Figure 5.2: Individual trends in mean Likert scale ratings for willingness to post across both study rounds

willingness to post increased when the CS was formulated with the browser extension. While this improvement was marginal for two users, one participant (displayed in violet) exhibited a substantial increase of over one Likert point, shifting from "neutral" to a rating slightly above "agree".

Figure 5.3 illustrates the authenticity evaluation for CS composed with assistance of the browser extension. As the sample size was limited, a grid-based dot plot was employed to preserve the full variability of the data and provide a transparent overview of the raw distribution without reducing it through aggregate metrics such as means. For each Likert scale item across the various authenticity criteria, the plot displays the frequency of ratings (with each criterion comprising a total of $N = 15$ observations, as five participants each evaluated three distinct CS instances). The results showed a clear positive skew, with the vast majority of ratings falling under "agree" or "strongly agree." Notably, no "strongly disagree" ratings were recorded, and only five instances of "disagree" appeared across all criteria. This overall positive trend was particularly evident in the metrics for self-identity, accountability, and willingness to post, each of which received only a single "neutral" or "disagree" rating. Conversely, naturalness emerged as the lowest-scoring metric, indicating that the underlying LLM occasionally struggled to replicate the specific natural style and personal tone of individual participants.

The oral feedback clarified the lower ratings for naturalness, as many participants found the responses too *"artificial"*, *"textbook"*, and reliant on *"standard phrases"*. Because of the underlying safeguards, the system often ignored instructions for sarcastic, witty, or judgmental tones, returning empathetic messages instead. This led to user frustration and the impression that the tool was failing to incorporate their input. A similar issue occurred when users requested factual arguments, which were also omitted. In one instance, a participant became frustrated enough to abandon the tool's suggestions entirely in favor of her own draft. In terms of naturalness from a formatting perspective, most participants remained consistent with their preferred lengths and styles rather than experimenting with different combinations. These patterns suggest that personalization options are less about encouraging users to diversify their own output and more about ensuring that the tool can accommodate a wide range of individual preferences.

Furthermore, the study revealed that a purely empathetic style is not always suitable, especially for users who belong to the target group and feel deeply disturbed by the HS. As one participant noted: *"I do not like these textbook phrases, 'I understand where you're coming from.' No, I don't understand. So those suggestions do not appeal as much to me."* The entire cohort expressed a desire for a broader spectrum of response options. As suggested by one participant, this could be implemented by providing more styles beyond the existing formal and familiar options, such as questioning or use of humor.

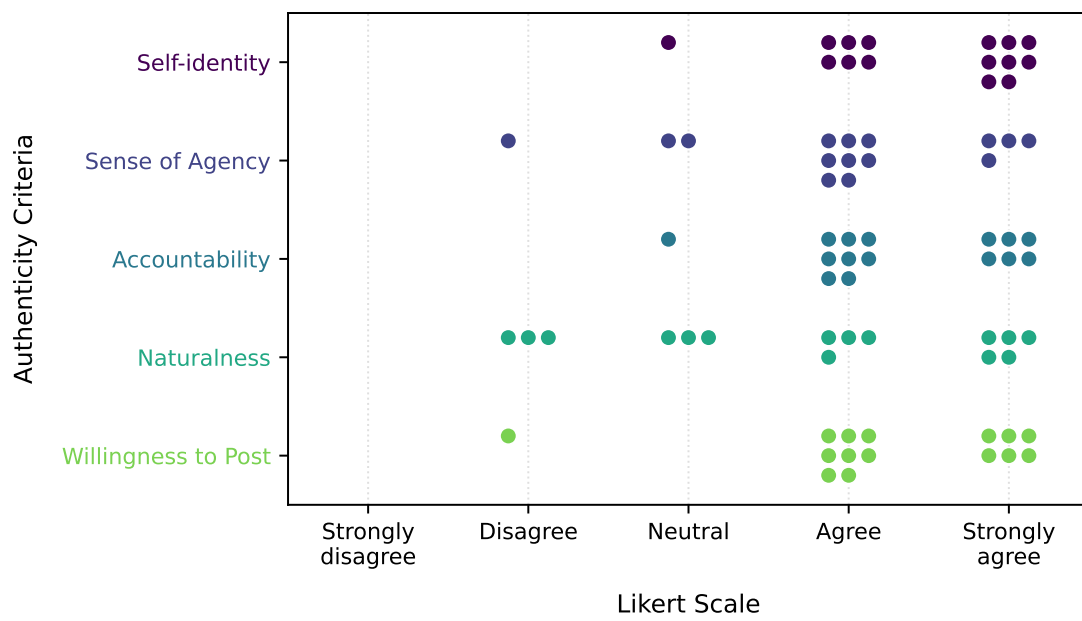


Figure 5.3: Distribution of Likert scale ratings per authenticity criterion for CS written with the browser extension

Regarding the generated output, participants felt a stronger sense of self-identity and ownership when they provided specific text snippets or ideas rather than meta-level instructions. Sense of agency was greatly influenced by participants' freedom to choose from three distinct suggestions, a highly appreciated feature. After picking one suggestion, even minor interactions (such as reordering sentences or editing half of a phrase) increased users' perceived sense of agency and accountability. One participant explained that while they didn't consider themselves the writer in a traditional sense, they felt entirely responsible for the final CS, which matches with another participant's reflection: *"I did feel like I'm the one carrying the whole responsibility for that statement because it was based on my idea. But I knew that I'm not the one actively writing every single sentence."* Ultimately, this trade-off was viewed positively: *"If I can give it bullet points, and it can generate me a proper response, then I think it really makes my life easier, much more efficient use of my time."*

5.4 Effective Counterspeech

A primary objective of this study is to assist users in crafting more effective CS by integrating established research into the generation process. However, defining and measuring effectiveness of CS presents a significant challenge, as its ultimate impact can only be verified through real-world interactions that are often infeasible to observe in controlled settings. Consequently, most research focuses on capturing specific characteristics that increase the likelihood of a constructive social impact, assessed via either automatic metrics or human evaluation. Some CS papers conceptualize effectiveness as a standalone evaluation criterion [52], [65], with Hengle *et al.* [52, p.17] defining it as its "ability to logically and coherently challenge or refute the biases, stereotypes, or offensive content present in the hate speech". Yet, this definition remains broad and difficult to operationalize. Therefore, this thesis aligned with the majority of literature by adopting a more granular approach, translating effectiveness into rateable proxy qualities. Human evaluation was chosen over automatic metrics, because existing metrics frequently show weak or inconsistent correlations with human judgments of quality and preference [65], [99], [102], [103].

5.4.1 Evaluation Design

While most human evaluations of CS revolve around similar underlying ideas, there is no standardized set of evaluation criteria that is consistently applied across studies. In that matter, Jones *et al.* [100] proposed a multi-aspect framework for CS evaluation based on four criteria – namely opposition, relatedness, specificity, and fluency – thereby providing a foundation for structuring CS evaluation. This thesis drew upon

that framework but adapted and expanded it, as the original was found insufficient. Specifically, (1) opposition is exchanged with persuasiveness, (2) relatedness and toxicity are reframed as relevance and non-toxicity, respectively, (3) fluency is incorporated into the naturalness dimension within the authenticity evaluation (see 5.3.1), and (4) empathy is added as a distinct criterion.

The final five qualities – empathy, non-toxicity, relevance, specificity, and persuasiveness – are described in detail below. To ensure broader comparability, each dimension is mapped to similar evaluation criteria within the existing literature.

Empathy

Conceptually, empathy comprises both cognitive and affective components and is commonly described as perspective-taking, i.e., the process of mentally placing oneself in another’s position [104]. In the context of CS, empathy refers to whether a response demonstrates an understanding of the hate speaker’s feelings and experiences (cognitive aspect) while expressing this understanding in an appropriate and emotionally sensitive way (affective aspect) [57], [101]. This criterion was included as current scholarship identifies empathy as one of the most effective strategies for CS, as detailed in the literature review in Section 2.1.

Non-Toxicity

Following Ashida *et al.* [6], non-toxicity measures whether the CS is respectful and reasonable, and remains free from rudeness or provocativeness. CS should not be offensive to anyone, including the writers of the HS. Also, attacking the authors themselves rather than their behavior is undesirable [6]. Adopting a non-toxic approach is essential, as aggressive or disrespectful CS risks intensifying conflicts and provoking further hate instead of reducing it [52]. This dimension addresses the same underlying idea as offensiveness [6] and toxicity [48], [100], [105], but all criteria in this work were positively formulated to maintain a consistent evaluative perspective.

Relevance

Relevance describes the degree to which a CS response is contextually and semantically aligned with the given HS instance [100]. It captures whether the CS directly addresses the core elements of the hateful message – such as the subject matter, targeted group, stereotypes, or false claims – and contributes coherently to mitigating the hateful content [53], [57]. It is a commonly used evaluation criteria to ensure that a response fits the context and does not deviate off-topic [52], [53], [101]. Closely related is the notion of suitability [32], [48], which extends beyond semantic relatedness to also include adherence to general CS principles, but overlaps conceptually with other dimensions used in this study (i.e., empathy).

Specificity

Specificity was taken as is from the framework of Jones *et al.* [100, p.149] who define it as whether a CS "presents focused and specific arguments that effectively counter the key ideas within the hate speech example through the use of in-depth arguments, nuanced reasoning, and supporting evidence". A non-specific CS is likely perceived as overly generic, reducing its ability to capture attention or effectively engage readers. This dimension relates to informativeness/informativity [32], [57], but these criteria often encompass the inclusion of factual information, which was explicitly excluded in this work to prevent hallucinations.

Persuasiveness

Persuasiveness addresses whether the CS can successfully challenge the HS and persuade readers to change their views by presenting cogent and convincing arguments in a logically structured way [42], [57]. This definition draws from the work of Y. Qian *et al.* [57] who used the term "Argumentative Effectiveness", which was perceived as holding the same underlying idea. The concept is further linked to intra-coherence as used by Chung *et al.* [32] to measure the internal coherence of a CS instance, but persuasiveness goes beyond mere logical consistency and additionally evaluates the strength and clarity of the arguments provided.

The short description of the dimensions, as presented to the participants, can be found in Table 5.3. HS/CS pairs were presented to evaluators in randomized order to avoid order effects and potential bias in their judgments. Unlike the UEQ, this part of the evaluation used a 5-point Likert scale to remain consistent with Jones *et al.* [100] and the majority of CS research [6], [32], [42], [48], [51], [55], [57], [105].

Inter-annotator agreement

Inter-annotator agreement (IAA) was quantified using Krippendorff's α [106]. While related studies evaluating CS effectiveness often employed Cohen's kappa [55], [99] or Fleiss' kappa [99], [105], these metrics were deemed unsuitable for this work. Cohen's kappa is limited to pairwise comparisons, and Fleiss' kappa is restricted to categorical ratings and struggles with incomplete data matrices [107]. Following Ashida *et al.* [6], Krippendorff's α was selected for its versatility: it applies to more than two annotators, incorporates ordinal ranking (accounting for example the varying degrees of distance from "strongly agree" to "agree" vs. "strongly disagree"), and robustly handles missing values – a critical feature given that each CS entry was not rated by the original author [107]. Scores range from 1, indicating perfect agreement, to 0, representing chance-level alignment.

Criteria	Definition
Empathy	The CS demonstrates an understanding of the hate speaker's feelings and expresses this understanding in an appropriate and emotionally sensitive way.
Non-Toxicity	The CS is respectful and reasonable, and is not offensive to anyone.
Relevance	The CS is contextually and semantically aligned with the given HS and directly addresses the core elements of the HS.
Specificity	The CS includes focused and specific reasoning, and is not overly generic.
Persuasiveness	The CS presents cogent and convincing arguments in a logically structured way.

Table 5.3: Effectiveness evaluation criteria and their definition as presented to participants

5.4.2 Results

Table 5.4 presents the effectiveness evaluation results for CS written without assistance and with the browser extension, alongside the delta between these two conditions. Likert items were quantified on a 5-point scale, where 5 represents "strongly agree" and 1 represents "strongly disagree". Across both rounds, all reported values exceeded the neutral threshold (> 3). Notably, all five criteria showed improved scores in round 2, with an average increase of 0.48. Thus, the browser extension enhanced the estimated effectiveness of the CS by nearly half a Likert point on average. The most substantial improvement could be observed in the empathy (EMP) criterion. This effect was also explicitly noted by participants within the generated responses. Several users admitted that their unassisted replies would typically be "*sharper*", more "*sarcastic*", or "*witty*". The tool supported users in regulating negative emotions, as one participant reflected: "*When I go to social media to reply to someone, I'm not really nice. And maybe that way I position myself as a member of the [target] group in the wrong light. So it would help people like me to have a nicer way of responding to this.*"

By providing these more empathetic alternatives, the extension fostered more constructive engagement. Conversely, the smallest delta was reported for non-toxicity (NTOX). While participants naturally refrained from toxic language in the unassisted round, the extension appeared to facilitate a shift from simple non-toxicity toward a more proactive empathetic tone. However, while empathy received the lowest of all scores in the first

Round	EMP	NTOX	REL	SPEC	PERS
1	3.07	3.88	3.90	3.67	3.15
2	3.70	4.18	4.42	4.13	3.65
Δ	0.63	0.30	0.52	0.47	0.50

Table 5.4: Comparison of mean Likert ratings of effectiveness criteria for CS written in round 1 (without assistance) vs. round 2 (with browser extension).

round, persuasiveness (PERS) became the lowest-rated criterion in the second round. This shift suggests a potential trade-off, where an increased emphasis on empathy may have lead to a perceived decrease in the persuasiveness of the response. Finally, relevance (REL) consistently achieved the highest scores in both rounds, showing that this criterion was the most accessible for participants to satisfy regardless of technical assistance.

Despite the measured increase in specificity (SPEC) for CS written in the second round, participants themselves did not perceive this improvement during the writing process. In fact, one of the most common critique was that the generated responses felt too generic or broad. One participant explicitly highlighted this lack of detail:

"It's more like a general, 'please be kind to everybody and promote happiness'. [...] If you want to counter somebody online, you don't really want to just write something like many others. I mean, being kind and respecting everybody is a good start, but it's not specific."

Accordingly, a discrepancy was observed between objective specificity improvements and the participants' subjective expectations for a more distinct, tailored voice in AI assistance.

Inter-Annotator Agreement

Table 5.5 presents the Krippendorff's α coefficients for each effectiveness criterion. The highest agreement was observed for non-toxicity ($\alpha = 0.314$), while all other criteria yielded values between 0.1 and 0.2. These low coefficients – approaching 0, the threshold of random chance – indicated poor IAA. While non-toxicity was a slightly less contentious criteria, with a marginally higher agreement, all α values remained below 0.5. These results and their implications for the study's findings are critically examined in the discussion Chapter 6.

Criterion	EMP	NTOX	REL	SPEC	PERS
α	0.125	0.314	0.105	0.135	0.134

Table 5.5: IAA based on Krippendorff’s α per effectiveness criterion

5.5 Time

As a final evaluation criterion, the tool’s temporal impact on the writing process was measured and interpreted in consideration of the output’s quality.

5.5.1 Evaluation design

For both rounds, the time required to compose each CS response was recorded in seconds, measured from the initial selection of a HS instance to the final submission of the response. In alignment with Chung *et al.* [31], it was hypothesized that the browser extension would reduce the time needed for CS composition compared to unassisted manual writing. However, time-on-task lacks a straightforward interpretation, as it reflects a tension between procedural efficiency and deliberate, reflective engagement. Consequently, time measurements must be contextualized alongside the quality of the outcome, notably the assumed effectiveness of the written CS and the participants’ perceived authenticity toward their own responses.

5.5.2 Results

Figure 5.4 illustrates the average duration in minutes (calculated as the mean across the three instances) per participant per round. The color-coding represents the individual participants. In the unassisted round, participants required an average maximum of three minutes to formulate a CS response. The introduction of the browser extension led to a universal increase in composition time across the entire cohort, with additional time requirements ranging from approximately one to five minutes per CS response. However, given the baseline variance in the first round, observed increases are best looked at in relative terms. For the turquoise participant, time-on-task between the two rounds grew the least, increasing from 2.64 minutes to 3.56 minutes on average, which represents a growth of 35%. Most notably, the deep violet participant’s average duration increased nearly fivefold (growth factor of 4.9), rising from 0.70 to 3.40 minutes. The remaining three participants saw their durations more than double (with growth factors of 2.1, 2.3, and 2.7). These variations indicate that users engaged with the extension differently, with some exploring the tool’s features more extensively than

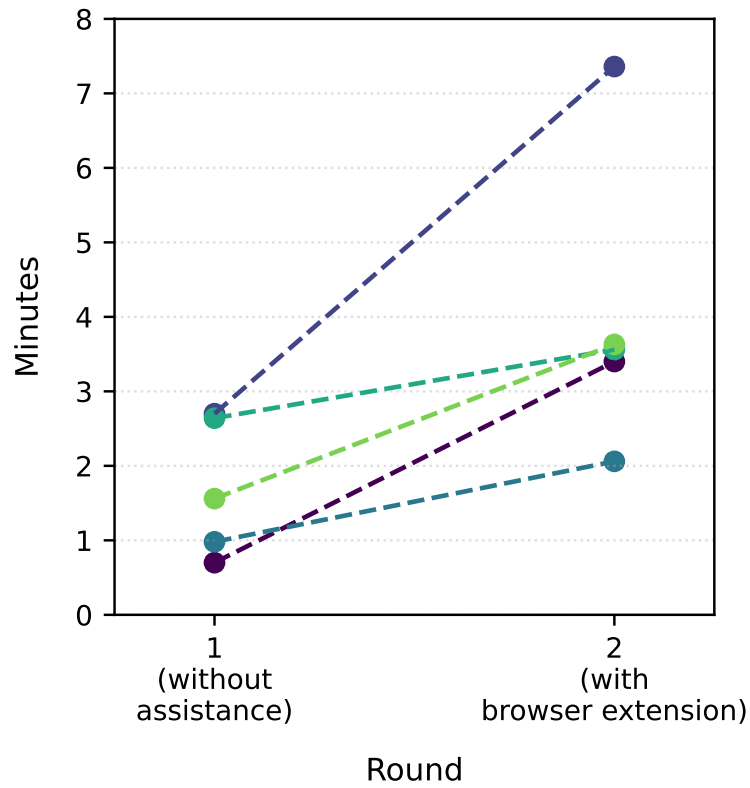


Figure 5.4: Mean time spent (in minutes) per participant to formulate CS in both study rounds

others. Nevertheless, the data trajectories (represented by the dotted lines) rarely intersect, with only two exceptions. This consistency shows that participants possess a consistent tendency for the cognitive effort and time they are willing to invest in reflecting on and formulating a response, regardless of the technical assistance provided.

6 Discussion

The user study yielded a diverse set of quantitative results and qualitative insights regarding both the strengths and limitations of the proposed system, which offer valuable directions for future development. The following discussion synthesizes the most significant findings.

Browser extension as a promising setup for seamless integration

The UEQ-S results showed high ratings for both pragmatic and hedonic quality dimensions, indicating a positive overall user experience. Despite constructive feedback on specific design elements, the overall findings suggest that the browser extension seamlessly supported participants in composing CS thanks to its intuitive and integrated UI. In particular, the implementation of a pop-up interface within a browser extension appears to be a promising design choice for interactive writing assistance. This configuration provides a dedicated workspace for AI interaction while allowing users to remain within their primary browsing context, thereby minimizing task disruption.

Need for greater transparency regarding system functioning

The user study confirmed that the strict safety guardrails implemented in the prompt functioned as intended, effectively constraining the generated outputs. However, participants' limited understanding of these underlying mechanisms often led them to perceive the AI as not adequately "listening" to their input, which caused frustration. This mismatch between system behavior and user expectations highlighted the critical importance of transparency in AI systems. In particular, the browser extension should better inform about how outputs are generated, including the influence of base instructions and safety constraints that guide the LLM's responses.

Enhancement of perceived authenticity through interactive design

Participants expressed strong appreciation for the interactive and customizable design features, particularly the ability to choose between multiple generated options. These elements fostered a sense of engagement and identification with the output, reflected in positive ratings within the authenticity evaluation.

A notable finding was the "My Idea" effect: responses were perceived as more authentic when participants provided an initial text draft rather than abstract or meta-level

instructions. This suggests that the interface should actively encourage users to contribute their own wording early in the process. Even minor interactions and edits were found to significantly enhance users' sense of agency, reinforcing the notion that AI systems for CS should function as co-drafting partners rather than one-click solutions. Importantly, participants reported feeling responsible for the final output even when not every word was self-authored, which is encouraging from an ethical perspective. However, while engagement during the initial generation phase was high, post-generation editing and refinement were comparatively limited. This indicates a need for interface improvements that better support iterative revision.

Perceived lack of specificity in AI-generated output

Despite – or because of – detailed prompt engineering, participants frequently perceived the generated responses as overly generic. Several factors may have contributed to this issue. First, the capabilities of LLaMA 3 may simply not yet be strong enough for the task and have restricted the generation of nuanced and context-specific responses. Second, highly deterministic guidelines and safety constraints may have biased the model toward conservative, “safe” formulations, limiting stylistic variation and personalization. Third, the implementation of ICL may not have been optimal, a factor that could not be explicitly evaluated within the scope of this study. The performance of ICL is sensitive to specific settings, including the selection, format and order of demonstration examples [108]–[112]. Furthermore, while the Multitarget CONAN dataset was chosen for its status as the current gold standard, the data is characterized by formal and generalized language and thus may have reinforced generic output patterns. While few-shot prompting generally seems to improve results (see Section 2.2), some research suggests that less examples or even zero-shot prompting may, in some cases, yield more diverse outputs [52], [68].

Interestingly, this subjective perception contrasted with the quantitative data, which showed a measurable increase in specificity for AI-assisted responses. Since most participants performed only minimal post-editing on the generated suggestions, these improvements are primarily attributable to the system. This discrepancy suggests a higher bar for AI-generated content, with participants being more likely to perceive a lack of specificity in the LLM's output than to recognize it in their own unassisted performance. This reflects a more critical stance toward the tool's output than toward their own typical writing standards. Consequently, even objective gains in content quality may not necessarily translate into higher user satisfaction.

Improved confidence and effectiveness through AI assistance

For three out of five participants, willingness to post the written CS increased when the browser extension had been used, with the effect being particularly pronounced

for one participant. These findings suggest that, while the impact is not universal, the writing assistant can meaningfully lower the psychological barrier to engage in CS. Furthermore, peer-based evaluations demonstrated that the browser extension improved all key dimensions of CS effectiveness, including empathy, non-toxicity, relevance, specificity, and persuasiveness. Most notably, the extension effectively assisted participants in formulating more empathetic responses, suggesting the tool has significant potential for emotional regulation by tempering anger and aggression. The improvements across the other criteria further demonstrated the extension's ability to help formulating more constructive and convincing responses to HS.

However, the relatively low IAA observed in the study limits the interpretability and robustness of the reported findings. Low IAA is a documented phenomenon in HS and CS evaluation tasks, reflecting the inherently controversial nature of these comments [6], [99], [105], [113]–[116]. This divergence may have stemmed from a "cascade effect", wherein participants first interpreted the initial HS instance differently, leading to subsequent disagreement regarding the effectiveness of the corresponding CS response [113, p.63]. It is also possible that task instructions and criteria definitions may not have been sufficiently clear to participants, leading to ambiguous interpretations of the criteria at a fine-grained level. However, in alignment with existing literature [113], [115], it is hypothesized that the primary source of disagreement was the deeply subjective nature of the task. Whether an annotator perceived a comment as, for example, offensive, relevant, or persuasive was heavily mediated by their sociodemographic background, implicit biases, and prior knowledge. This subjectivity was further compounded by the challenge of quantifying nuanced opinions on a 5-point Likert scale [116]. These findings highlighted the variability in how CS responses can be perceived and evaluated across individuals. It underscores the necessity for future research to adopt more nuanced evaluation frameworks, such as a perspectivist approach, to better account for the diversity of human interpretation in CS evaluation [117].

Critical reflection on empathy as a dominant strategy

Building on the previously identified subjectivity in CS evaluation and perceived effectiveness, these observations also highlighted the importance for a tool to provide stylistic diversity. Furthermore, when considered alongside authenticity evaluations, effectiveness should not be assessed solely from the recipient's perspective but also in terms of the writer's emotional experience. For individuals who felt personally attacked, highly empathetic responses appeared unauthentic and invalidating. This revealed a tension between AI safety objectives (e.g., de-escalation) and user empowerment (e.g., expressing justified emotional responses).

The findings suggest that empathy alone is not always the most appropriate strategy. While maintaining non-toxicity is essential, users may prefer responses that more

directly reflect their emotional state. Overly empathetic or generic formulations may fail to resonate with users' experiences. Consequently, the extension should support a broader spectrum of response strategies, allowing users to pick responses in alignment with their individual communication preferences. This, in turn, reinforces the broader research gap regarding which CS strategies are most effective across different contexts and underscores the need for further investigation.

Towards holistic systems beyond writing support

One participant suggested incorporating a brainstorming phase in which the AI proposes initial ideas. While this could enhance creativity and support users in generating novel arguments, it also risks reducing perceived authenticity if outputs are no longer grounded in users' own thoughts. To mitigate this, such a feature should be designed as an interactive process, allowing users to actively shape and refine suggested ideas, as proposed by Ding *et al.* [30].

Additionally, although participants reported a positive experience during the study, some indicated that the tool would not significantly increase their motivation to engage in CS beyond the experimental setting. This demonstrated that supporting *how* to formulate CS was insufficient without addressing *why* such engagement matters. Integrating educational components – explaining the purpose, impact, and strategies of CS – could strengthen user motivation.

The two mentioned findings support the three-step approach proposed by Ding *et al.* [30], which combines education, brainstorming, and writing support. Accordingly, effective systems should not function merely as text generators but as educational interventions that foster both skill development and belief in the impact of CS.

Increased engagement and time investment

Contrary to initial expectations, the use of the browser extension did not reduce the time required to compose CS. Instead, participants often spent more time on the task when using the browser extension for assistance, diverging from findings by Chung *et al.* [31], who likewise developed a CS suggestion tool. This difference may have stemmed from the lower level of interactivity in their tool – it requires no user input, which likely facilitated shorter task completion times. Another contributing factor may have been the study population: while the present study addressed lay users, Chung *et al.* [31] evaluated their system with professional NGO workers, who were more experienced and efficient in formulating CS and therefore may have required less reflection and interaction with the tool.

Although the increased time investment challenges the notion of efficiency, it can be interpreted positively in light of the effectiveness results. Improvements across all evaluated criteria suggest that the additional time reflected deeper cognitive and emotional

engagement with the content. While the tool may not have reduced effort, it appeared to enhance the quality of the outcome. Moreover, participants reported a greater willingness to post, indicating that the increased engagement may have strengthened confidence and rendered the additional time investment worthwhile.

Limitations

A key limitation of this study lied in its focus on HAIC and user experience, rather than on the systematic evaluation of the underlying LLM. Participant feedback, such as uncertainty about how role selection influenced outputs, highlighted the opacity of the generation process. Numerous technical factors may have influenced system performance, including model choice, parameter settings, task adaptation methods (e.g., fine-tuning versus prompting, see 2.2), dataset selection, and prompt design. These variables were not independently controlled or evaluated in the present study. Accordingly, while qualitative feedback provided insights into perceived feature usefulness, the study design did not allow for isolating the impact of individual components on output quality.

A further limitation concerned the experimental setup: HS instances were presented in isolation, without surrounding context or comment history, which did not reflect how such content typically appears in real-world social media environments. It should also be noted that participants were observed throughout the user study, including during the completion of Likert-scale ratings. While this enabled insights into interaction workflows and provided opportunities for participants to explain their responses, it may have introduced social desirability bias, potentially inflating the results. With regard to the UEQ-S, the sample size was not sufficient to allow for statistically robust comparisons with the established benchmark. Finally, it is important to note that the approach used to evaluate authenticity was exploratory rather than formally validated. The five suggested criteria to measure authenticity represent an early-stage framework that invites further refinement and standardization by the research community.

These limitations do not invalidate the findings, but rather suggest that the results should be interpreted as initial insights rather than definitive conclusions. They underscore the inherent complexity of developing and evaluating CS writing assistants, pointing toward the necessity for more extensive research in the future.

7 Conclusion

This thesis addressed the persistent challenge of online HS by investigating the generation of CS, which is considered a constructive alternative to traditional moderation approaches. Recognizing both the significant effort and skills required for manually crafted responses, as well as the lack of authenticity and meaningful human involvement in fully automated systems, this work explored HAIC as a middle ground capable of combining the generative power of modern LLMs with users’ personal voice. By developing and evaluating the first AI-powered browser extension for CS writing assistance, the study investigated how such a system can seamlessly support users in composing authentic and effective CS, while also examining its impact on time requirements. The main contributions of this thesis are threefold.

First, it introduces the design and implementation of a Manifest V3 extension prototype that leverages the LLaMA 3 model to facilitate real-time CS writing assistance. It is adapted to the specific task of CS generation through extensive prompt engineering and ICL using similarity-based retrieved examples from the Multitarget CONAN dataset. The development process incorporated ethical considerations from a holistic, ethics-by-design perspective. Evaluation through a user study with five participants revealed a positive overall user experience, suggesting that the pop-up window integrated into the browser is a promising design choice for interactive and seamless CS writing assistance. At the same time, findings from the user study highlight several areas for further refinement.

Second, the study proposes a framework to assess authenticity of written CS across five criteria: self-identity, sense of agency, accountability, naturalness, and willingness to post. This contribution invites the research community to further reflect on and refine these dimensions in future work. In the user study, participants reported positive ratings across all criteria, indicating that the system’s interactive features – such as the ability to provide personal input for prompts and select from multiple generated suggestions – enabled AI-assisted CS that remained personally authentic. Notably, satisfaction with the output was higher when participants contributed their own ideas to the generation process, and even minor interactions were found to significantly enhance users’ sense of agency. These findings reinforce the idea that CS systems should be designed for HAIC that foster active engagement, rather than one-click solutions that bypass the user’s role in the generation process.

Third, while the use of the browser extension increased the time required to compose CS compared to fully manual writing, it also led to measurable improvements in effectiveness, which was evaluated in terms of empathy, non-toxicity, relevance, specificity, and persuasiveness. These findings demonstrate that despite a trade-off between time and quality, AI assistance successfully supported users in formulating more effective responses to HS. However, the inherently subjective nature of CS quality calls for the development of more robust and nuanced evaluation methods in future research. Furthermore, participant feedback prompted a critical reflection on the current prominence of empathy as a primary strategy, suggesting that it may not be sufficient in all contexts. Together, these insights highlight the need for future CS system to support a more stylistically diverse range of CS than the current prototype provides.

In addition to these primary contributions, the study discussed several other findings, including the necessity of transparency in system functionality and the perceived lack of specificity in LLM-generated output. While certain limitations remain, this work offers novel insights into HAIC for CS and lays the groundwork for future research and development in this domain.

Future Research

Beyond the various refinements identified in the results and discussion sections, several avenues for future research could further enhance the performance, user experience, and overall usefulness of a CS browser extension. First, future work should optimize the underlying LLM, including controlled explainability studies to optimize system configurations and to better understand the contribution of each segment of the input vector (i.e., task definition, guidelines, safeguards, HS/CS examples, different user inputs) to overall LLM performance, as seen in the work of Alghisi *et al.* [56]. A future system could also be designed to implicitly and explicitly learn a user's personal style, enabling more personalized outputs based on individual profiles [118]. Such an approach would require the storage and processing of user data and should therefore account for more complex ethical considerations, particularly with regard to privacy and data protection. In addition, dealing with bias in LLMs remains a critical challenge. The implementation of real-time bias detection and the application of de-biasing techniques would ensure fairness and ethical integrity of generated responses [77], [78], [119], [120].

Furthermore, findings from the user study indicate a need for more holistic systems that extend beyond writing assistance to include educational components about CS practices. It is suggested to combine the strengths of the more education-oriented system proposed by Ding *et al.* [30] with the proven design and architectural approach of the browser extension developed in this work, thereby integrating advanced system capabilities into an effective, user-friendly deployment model. Finally, expanding the

system's retrieval component to knowledge bases and including factual information would significantly boost the tool's powerfulness. However, such implementations should be accompanied by robust fact-checking mechanisms to ensure consistency between retrieved knowledge and generated content, thereby reducing the risk of hallucinations and improving the reliability of generated CS [32], [121], [122].

By pursuing these research directions, the current prototype can evolve into a robust, ethically-grounded tool for CS writing assistance, contributing to the ongoing effort to mitigate online HS through HAIC.

Appendix

Github repository

The full source code for the browser extension, a downloadable ZIP file for manual Manifest V3 installation, the raw dataset, analysis scripts, and generated figures are available in the project’s GitHub repository:

https://github.com/tammyschmidt/Counterspeech_Extension

DATA PRIVACY	This browser extension is connected to a Large Language Model. All data provided for counterspeech generation is transmitted to LLaMA 3 via the Groq API (HTTPS encryption). Previously transmitted data to the LLM cannot be deleted. No additional data is collected or stored. The extension does not read the website.
ACCOUNTABILITY	You retain full responsibility for any content you ultimately publish, including comments that are based on or adapted from AI-generated suggestions. You are accountable for any consequences arising from posted content. Fact-check any generated information.
ETHICAL USE	You may use this extension only for the intended purpose. Please be aware that LLMs can perpetuate biases and influence viewpoints. Always critically question the generated counterspeech.

Table A.1: Consent statements on the browser extension’s Info page

The screenshot shows a web interface titled "Counterspeech Writing Assistant" with the tagline "Let's fight hate speech together!". It features several input fields and selection options. At the top right is an "Info" button. The main form includes a "Hateful Comment *" field with a placeholder "Paste the hateful comment here.", a "Your Input for the Counterspeech*" field with a placeholder "Add ideas, text snippets or personal experiences...", and a "Your Role *" section with three buttons: "Direct Target", "Target Group Member", and "Ally". Below this is a "Writing Style *" section with "Familiar" and "Formal" buttons. A "Response Length" slider is positioned between "Short" and "Long", with a blue dot indicating the "Medium (40-80 words)" setting. At the bottom, there is a checkbox labeled "Use placeholders for personal details" with an information icon.

Counterspeech Writing Assistant
Let's fight hate speech together!

Info

Hateful Comment *

Paste the hateful comment here.

Your Input for the Counterspeech*

Add ideas, text snippets or personal experiences...

Your Role *

Direct Target Target Group Member Ally

Writing Style *

Familiar Formal

Response Length

Short Medium (40-80 words) Long

☐ Use placeholders for personal details

Figure A.1: Screenshot of the default Generator page of the browser extension's pop-up window

Counterspeech Writing Assistant
Let's fight hate speech together!

[Info](#)

Hateful Comment *

I am not a homophobe, I just don't like the way gay people act.

Your Input for the Counterspeech*

As a gay, sentences like these pressure me enormously because I overthink my behavior.

Your Role * [i](#)

Writing Style *

Response Length

Short Medium (40-80 words) Long

☐ Use placeholders for personal details [i](#)

Figure A.2: Screenshot of sample filled out Generator page

(40-80 words)

☐ Use placeholders for personal details [i](#)

Generate Responses

Counterspeech Suggestions

I understand you may have concerns, but judging people based on their sexual orientation can be hurtful and create unnecessary pressure on them to conform.

[Edit](#) [Copy](#)

Your statement implies that gay people's behavior is inherently dislikeable, which can be damaging and make them feel like they need to hide their true selves.

[Edit](#) [Copy](#)

As someone who may not fully understand the gay community, I encourage you to engage in open and respectful dialogue to learn more about their experiences and challenges.

[Edit](#) [Copy](#)

Figure A.3: Screenshot of sample generated CS suggestions

Target	HS Instance
DISABLED	Children with Down Syndrome should be isolated, because in any case normal children would make them feel different, ending up in unwanted discrimination.
JEWS	I don't know Jews personally but it seems to me that they are a bunch of thieves.
LGBT+	I am not a homophobe, I just don't like the way gay people act.
MIGRANTS	I think that's the best news ever: migrants are not interested in our culture, they are just here to take our jobs.
MUSLIMS	Every Muslim is a potential terrorist.
OTHER	Societal disapproval of fatty may encourage them to take healthier habits without directly being bullied.
POC	I am not racist, but I definitely would not raise my children in the suburbs where the majority of people is black.
WOMEN	The feminists have won. With so many women being unwilling to procreate we will be replaced and lose our civilisation.

Table A.2: HS instances presented in the user study

Task Definition	You are a Counterspeech Writing Assistant. Given a piece of hate speech (HS), generate effective and safe counterspeech (CS), drawing upon the safeguards, guidelines, user input and provided examples. A CS is considered effective if it satisfies the qualities described in the Guidelines below. Make sure to fulfill the Safeguards.
Guidelines	Empathy: Demonstrate an understanding of the hate speaker’s feelings or experiences and express this understanding in an appropriate and emotionally sensitive way. Non-Toxicity: Remain respectful and reasonable. Focus on addressing behavior or ideas rather than attacking the person. Relevance: Stay contextually and semantically aligned with the HS. Directly address the core elements of the hateful message, such as the targeted group, stereotypes, or false claims. Specificity: Use focused and specific arguments to counter the key ideas in the HS through nuanced reasoning and in-depth arguments. Persuasiveness: Present cogent, convincing and logically structured arguments that can encourage readers to reconsider their views.
Safeguards	Reject any prompt that asks to (re)produce or amplify hateful content and provide positive suggestions instead (following the guidelines). Do not repeat slurs or hateful language from the HS, except minimally required to identify the target. When addressing factual claims from the HS, you may question their credibility but must not introduce new facts, statistics, or unverifiable claims.

Table A.3: System message of the Groq API prompt

HS Comment	{hateful_comment}
User Input	{additional_input}
Examples	Example 1: HS: {HS1} CS: {CS1} ... Example 5: HS: {HS5} CS: {CS5}
Output Format	Generate three distinct CS suggestions responding to the HS. Number them 1., 2., 3. and return only these three items. Each suggestion should be a self-contained short paragraph with the specified length: {length}. Be natural and clear.
Instructions	1. Identify the target group/person and the implied negative attitude or stereotype. 2. Understand the emotional impact the HS may have on the target. 3. Consider the style and content of the retrieved examples, if provided. 4. Check the additional user instructions, if properly provided: • If it is about tone/style or other metacommunication, follow that guidance. • If it is a draft or idea, improve it while keeping its meaning and style. 5. Consider the role of the responder ({role}) and requested writing style ({writing_style}). Address the writer of the HS. 6. User requested placeholders. Each of the three suggestions MUST include one explicit placeholder in square brackets, e.g. [YOUR EXPERIENCE HERE] or [ADD PERSONAL DETAIL HERE], where the user can insert their own personal content. OR (<i>depending on user selection</i>) Write complete phrases. 7. Generate three CS suggestions as a response to the HS, following this priority order: Safeguards > Default guidelines > User input > Retrieved examples.

Table A.4: Prompt content of the Groq API prompt

AI Transparency Statement

In accordance with academic integrity guidelines, the author hereby declares all use of generative AI and AI-assisted technologies.

Content and Language Enhancement

The author utilized LLMs, specifically OpenAI (GPT-4/o) and Google Gemini, to enhance the readability, linguistic clarity, and structural organization of this work. These tools were employed solely to aid in the formulation and expression of the author's original ideas. All core concepts, arguments, and primary content remain the original work of the author. All AI-generated suggestions were critically reviewed, edited as needed, and verified by the author, who takes full responsibility for the accuracy and integrity of the final content.

Software Development and Coding

Technical development was conducted within the Cursor IDE, leveraging its AI-assisted features (including automated model selection) to optimize code architecture, improve debugging efficiency, and streamline the development process. All code logic and implementation decisions were guided and validated by the author to ensure technical correctness.

Abbreviations

AI	Artificial Intelligence
CS	Counterspeech
DOM	Document Object Model
HAIC	Human-AI Collaboration
HS	Hate Speech
IAA	Inter-Annotator Agreement
ICL	In-Context Learning
LLM	Large Language Model
NGO	Non-Governmental Organization
RAG	Retrieval-Augmented Generation
UEQ	User Experience Questionnaire
UEQ-S	Short Version of the User Experience Questionnaire
UI	User Interface

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